# ProtCOVER: Family-Aware Conformal Prediction for Variant Effect Predictors on Unseen Protein Families

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**Abstract.** Protein language models and related variant effect predictors (VEPs) increasingly classify missense variants in proteins evolutionarily distant from those used for calibration. Split conformal prediction over variants is valid for its own target, a random variant dominated by well-studied families, but deployment needs coverage for a variant of a new family. On ProteinGym clinical substitutions with five zero-shot predictors, Mondrian conformal comes within 0.005 of that target for four predictors, yet family-weighted pathogenic coverage is only 0.823–0.873 at a 0.90 target for all five, without any shift. We present ProtCOVER, a conformal procedure that treats homology families, rather than individual variants, as the exchangeable unit. ProtCOVER (i) calibrates at the family level over label-free homology clusters, giving marginal coverage for a new family under family exchangeability; (ii) adapts to descriptor shift by calibrating conditionally on an estimated density ratio of label-free descriptors (sequence similarity, alignment depth and embedding distance; a variant reweights by the ratio instead), with finite-sample family-weighted coverage, valid as computed, whenever the true ratio is a nonnegative affine function of the estimate; and (iii) can abstain at low effective calibration support, which proved no better than random abstention. Under a shift of its descriptors, ProtCOVER's mean worse-class shortfall below the target is 0.000, against 0.010 for group-weighted and 0.029 for hierarchical conformal prediction, with larger sets (about equal at matched coverage); on orphan families and held-out embedding regions it matches group weighting (0.002 against 0.003). In evaluations following pre-registered protocols, every shift-aware method met the target on new ClinVar variants of benchmark proteins, which ESM-1b separated better; on 12,225 proteins outside the benchmark, every method fell short for pathogenic variants with either of two predictors, ProtCOVER and a variant conditioned on the descriptors the least, within the 0.01 margin of group weighting.

**Keywords:** conformal prediction · variant effect prediction · protein language models · uncertainty quantification · distribution shift · selective prediction · ProteinGym · protein families

## 1 Introduction

Zero-shot variant effect predictors (VEPs), from protein language models such as ESM-1b [38,9] to alignment-based models such as EVE and GEMME [16,23] and hybrids such as TranceptEVE and PoET [33,42], now rank missense variants with high accuracy on clinical benchmarks [32]. Clinical use, however, needs more than a ranking: variant interpretation under the guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) [37] needs evidence whose error rate is known, and the ClinGen calibration of computational evidence [35] is genome-wide, while a single genome-wide calibration can hide large differences between genes [40].

Conformal prediction [44,1] offers distribution-free error control: a set over $\{\text{benign}, \text{pathogenic}\}$ that contains the true label with probability at least $1 - \alpha$. Its guarantee rests on exchangeability between calibration and test data. For VEPs this assumption fails in two ways. First, variants are grouped into proteins and homologous protein families: a predictor's accuracy differs from family to family, and families contribute very different numbers of labeled variants (from 2 to 741 per protein in our data). Pooling variants lets the largest families set the error rate, so the "90%" is a statement about variants from well-studied genes, not about a new protein. Second, the proteins a predictor is deployed on are often evolutionarily distant from, or have shallower alignments than, the proteins in the calibration set: a family-level distribution shift that is visible without labels. We address both with ProtCOVER, an uncertainty-quantification layer on top of a predictor's scores that calibrates over homology families instead of variants: its output is a prediction set with controlled error, not a variant classification or an ACMG/AMP evidence strength.

We make three contributions.

- *Problem and evidence.* We formalize *family-weighted* coverage, the probability that a random variant of a new protein family is covered, and its class-conditional version for pathogenic variants, and show that variant-level calibration, which is valid for its own target, misses this one. On ProteinGym clinical substitutions, class-conditional (Mondrian) split conformal, the standard remedy for class imbalance, comes within 0.005 of its own variant-level target for four of five predictors yet covers only 0.823–0.873 of pathogenic variants per family for all five at a 0.90 target, with no distribution shift at all; pre-registered tests confirm this on ClinVar variants added after the benchmark (0.873 for ESM-1b) and on proteins outside it (0.778). Family-level calibration removes the failure when families are exchangeable.
- *Method and guarantees.* ProtCOVER calibrates at the family level and conditionally on a cross-fitted density ratio of label-free family descriptors; an optional gate can abstain when calibration support is low. Family-level pooling is not new: hierarchical conformal prediction (HCP) [25] is conformal risk control [2] with families as units. Nor is the form of our program: it is the conditional risk control of Blot et al. [7], which carries the conditional calibration of Gibbs et al. [18] to such losses. What Theorem 1 adds is a family-level construction that is valid in finite samples *as computed*: the test family's labels are unknown, so its scores sit at $+\infty$; the minimizer need not be unique, so the threshold is the largest optimal fitted value; the program can be unbounded, so the set is then $\{0, 1\}$; and each computed threshold is certified conservatively in exact arithmetic. The resulting family-weighted quantile regression has family-weighted coverage at least $1-\alpha$ under every family-level shift whose density ratio is a nonnegative function in the span of its features, and gives HCP's threshold for constant features (up to ties). With the estimated ratio as a feature, ProtCOVER is valid under family exchangeability and when the estimated ratio is correct, and its loss under any other shift is bounded by the smaller of the bounds for HCP and for reweighting (Corollary 1).
- *Evaluation, including where the method does not help.* We test five predictors on five kinds of split (exchangeable; shifted descriptors; orphan families and held-out embedding regions, both defined without the descriptors; and a shift in predictor accuracy that violates our assumptions), with robustness checks of the base model, the descriptors and the family definition, and two external evaluations that follow pre-registered protocols, one on 12,225 proteins outside the benchmark with two predictors. Under descriptor shift ProtCOVER has the smallest worse-class shortfall below the target; on orphan families and held-out regions it matches group weighting, and it never exceeds a baseline's shortfall by more than 0.01, at the price of larger sets under shift. When the predictor's accuracy shifts, in a stress test and on the new proteins, every method under-covers, ProtCOVER's conditional versions least but within the margin of

group weighting; the temporal evaluation does not separate the shift-aware methods; and the optional gate is a negative result: its support-based abstention does not beat random abstention.

## 2 Related Work

*Conformal prediction.* Split conformal prediction [44,26] gives marginal coverage under exchangeability, and Mondrian versions calibrate within categories [43,12]. Under covariate shift, weighted conformal prediction uses a known or estimated likelihood ratio [41,27], related methods handle label shift, non-exchangeability, proximity and multiple sources [36,4,20,10,29], and group-weighted conformal prediction (GWCP) [5], our main baseline, handles shifts in the proportions of known groups. Conditional calibration gives coverage over a class of covariate shifts by quantile regression with the test point imputed [18], related methods give group-conditional coverage [22,3], and conformal risk control and its conditional version extend these to bounded monotone losses [2,7], all for exchangeable points. For grouped data, Dunn et al. [14] proposed pooling and subsampling for a new group, Lee et al. [25] proved finite-sample guarantees for pooling (HCP), and later work uses a few observations from the test group [30]; Duchi et al. [13] calibrate one coverage threshold per environment, and macro-coverage [6] averages coverage over classes, in the spirit of our family weighting. In protein science, conformal methods have handled the feedback covariate shift of sequence design [15] and mining protein databases for function [8]. Table 1 compares the guarantees.

*Calibration of variant effect predictors.* ClinGen calibrated computational scores to evidence strengths for the codes PP3 (supporting pathogenicity) and BP4 (supporting benignity) [35]; later work found that genome-wide calibration masks heterogeneous performance across genes [40] and proposed gene- and domain-aware calibration [11], and RaCoon calibrates ESM-1b at the residue level [34]. These improve point probabilities within subgroups seen in training, without distribution-free guarantees, and condition on gene, domain or residue attributes rather than on homology support relative to the calibration data. Kermut [19] provides calibrated uncertainty for supervised variant effect regression on deep mutational scans. Selective classification [17] motivates our risk-coverage evaluation of abstention.

## 3 Problem Setup

*Families and scores.* A protein *family* is a cluster of homologous wild-type sequences obtained without labels (Section 6.1). Family $k$ is the random object $G_k = (D_k, N_k, (X_{k,i}, Y_{k,i})_{i=1}^{N_k})$, where $X_{k,i}$ describes a missense variant, $Y_{k,i} \in \{0,1\}$ is its label (1 = pathogenic), and $D_k \in \mathcal{D}$ is a *label-free descriptor* of the family: any function of its sequences, alignments, predictor scores, and of the fixed set of training families, but not of labels. A predictor score is turned into $\hat{p}(y \mid x)$ by a base model fitted on separate training families, which we treat as fixed. The nonconformity score is $s(x,y) = 1 - \hat{p}(y \mid x)$ and $S_{k,i} = s(X_{k,i}, Y_{k,i})$. We observe calibration families $G_1, \ldots, G_K$ with labels and a test family $G_{K+1}$ whose descriptor $D_{K+1}$ and variants $X_{K+1,i}$ are observed but whose labels are not.

*Coverage targets.* The unit of sampling is the family: a new family is drawn from the family-level law (of $G_{K+1}$), and the question is how often its variants are covered. For a set-valued predictor $C : \mathcal{X} \to 2^{\{0,1\}}$, the per-family coverage of the test family is $\gamma_{K+1}(C) = N_{K+1}^{-1} \sum_i \mathbf{1}\{Y_{K+1,i} \in C(X_{K+1,i})\}$, and our primary target is *family-weighted coverage*

$$\mathrm{Cov}(C) = \mathbb{E}\big[\gamma_{K+1}(C)\big] : \tag{1}$$

draw a new family, then one of its variants uniformly at random; Cov is the probability that this variant is covered. Every family counts once, whatever its number of variants. For example, among ten test families, one of 500 variants and nine of 5, the large one carries over 90% of the weight in variant-weighted coverage, which pools the variants (reported, not targeted), but a tenth in Cov. Its class-conditional version for label $y$ is $\mathrm{Cov}_y(C) = \mathbb{E}\big[N_{K+1,y}^{-1} \sum_{i:Y_{K+1,i}=y} \mathbf{1}\{y \in C(X_{K+1,i})\} \,\big|\, N_{K+1,y} > 0\big]$, where $N_{k,y}$ counts label-$y$ variants of family $k$; it averages over the families that have label-$y$ variants, not over all families. For a predictor used to flag pathogenic variants, $\mathrm{Cov}_1$ (pathogenic coverage) is the quantity of interest: $1 - \mathrm{Cov}_1$ is the family-weighted *pathogenic miss rate*, the rate at which a pathogenic variant's set excludes the pathogenic label. Coverage of each family cannot be guaranteed without assumptions on how families differ [43].

*Assumptions.* We use two alternative assumptions; each result states which one it needs. **A1** (*Family exchangeability.*) $(G_1, \ldots, G_{K+1})$ is exchangeable.

**A2** (*Family-level covariate shift.*) $G_1, \ldots, G_K \overset{\text{iid}}{\sim} P$ and, independently, the test family $G_{K+1}$ and any further unlabeled target families are i.i.d. from $Q$, where $Q \ll P$ and $\frac{dQ}{dP}(G) = w(D)$ for a measurable $w : \mathcal{D} \to [0, \infty)$. Equivalently, the descriptor's law under $Q$ is absolutely continuous with respect to its law under $P$, and the conditional law of a family's variants and labels given its descriptor is the same under $P$ and $Q$: only the distribution of descriptors shifts.

A1 holds when calibration and test families are drawn by a random split of a pool of families, regardless of dependence between families. It does not require exchangeability of variants within a family, because Cov averages over all variants of the test family (this is weaker than the hierarchical exchangeability of [25], which targets one variant of the new group). A2 formalizes deployment on families that differ from the calibration families in label-free ways, such as being evolutionarily distant or having shallow alignments; it fails when the predictor's accuracy shifts in a way that the descriptors do not see. Our shifted splits (S2–S5, Section 6.2) draw test families without replacement from one finite pool, and S3–S5 select them by properties (orphan status, embedding region, predictor accuracy) that the descriptors do not determine. A2 therefore holds at best approximately in S2 and not in S3–S5, where the distances of Corollary 1 can be close to one, so those splits only measure robustness; the simulations of Appendix C check the case the theory covers.

## 4 ProtCOVER

ProtCOVER has four components: (i) family-level calibration; (ii) adaptation to descriptor shift; (iii) an optional support gate; and (iv) class-conditional versions of each. Its default (ii) is conditional calibration with features $(1, \hat{w})$, for an estimated density ratio $\hat{w}$. Two variants replace it: *ProtCOVER-W* reweights the families by $\hat{w}$, and *ProtCOVER-D* conditions on the descriptors instead of $\hat{w}$.

*(i) Family-level calibration.* Let $\hat{F}_k = N_k^{-1} \sum_i \delta_{S_{k,i}}$ be the empirical distribution of calibration scores of family $k$. Given nonnegative family weights $\omega_1, \ldots, \omega_{K+1}$, set $p_k = \omega_k / \sum_{j=1}^{K+1} \omega_j$ and

$$\hat{\mu} = \sum_{k=1}^{K} p_k \hat{F}_k + p_{K+1}\, \delta_{+\infty}, \qquad \hat{q} = \text{Quant}_{1-\alpha}(\hat{\mu}), \qquad C(x) = \{y \in \{0,1\} : s(x,y) \le \hat{q}\}, \tag{2}$$

where $\text{Quant}_\tau(\mu) = \inf\{u : \mu((-\infty, u]) \ge \tau\}$. With $\omega \equiv 1$ every family, calibration or test, has the same total weight, spread evenly over its variants. The test family's weight sits at $+\infty$ because its labels, and hence its scores, are unknown: the guarantee holds for the threshold computed with its true scores, and moving them to $+\infty$ can only raise the threshold. This is the hierarchical conformal prediction (HCP) threshold of [25], which equals the CDF pooling of [14] up to the level adjustment.

*(ii) Adapting to descriptor shift: conditional calibration on an estimated density ratio.* Equation (2) uses one threshold for every test family. ProtCOVER lets the threshold depend on the test family's label-free descriptor through a feature map $\Phi : \mathcal{D} \to \mathbb{R}^p$ with $\Phi_1 \equiv 1$, spanning $\mathcal{F} = \{\Phi(\cdot)^\top b : b \in \mathbb{R}^p\}$. With $\tau = 1 - \alpha$ and the pinball loss $\rho_\tau(u) = \max\{\tau u, (\tau - 1)u\}$, let

$$L(b) = \sum_{k=1}^{K} \frac{1}{N_k} \sum_{i=1}^{N_k} \rho_\tau\big(S_{k,i} - \Phi(D_k)^\top b\big) - \tau\, \Phi(D_{K+1})^\top b, \qquad \hat{q} = \sup\{\Phi(D_{K+1})^\top b :\ b \in \arg\min L\}, \tag{3}$$

with $\hat{q} = +\infty$ if $L$ is unbounded below or if the fitted value $\Phi(D_{K+1})^\top b$ is unbounded over $\arg\min L$, and $C(x) = \{y : s(x,y) \le \hat{q}\}$. This is a family-weighted quantile regression in which, as in (2), each calibration family has total weight one and the test family's scores sit at $+\infty$ (the last term); with $\Phi \equiv 1$ it is the HCP threshold up to ties (Appendix A). $L$ is bounded below iff $-\tau\Phi(D_{K+1}) = \sum_j a_j \Phi(D_{k(j)})$ for some weights $(\tau - 1)/N_{k(j)} \le a_j \le \tau / N_{k(j)}$ on the calibration scores $j$, of families $k(j)$; otherwise the query's features cannot be balanced by the calibration families, for example when its estimated ratio far exceeds theirs, and

the set is $\{0,1\}$: a built-in abstention. The default feature map is $\Phi = (1, \hat{w})$, where $\hat{w}$ is an estimate of the density ratio $w = dQ/dP$ of descriptors, so the threshold can rise for families that are rare among the calibration families and frequent among the target families. Of the two variants, *ProtCOVER-W* instead reweights the families in (2) with $\omega_k = \hat{w}(D_k)$, $k \le K+1$ (computable because $D_{K+1}$ is label-free), and *ProtCOVER-D* uses the standardized descriptors other than length as features and needs no target families.

The ratio comes from a logistic classifier separating calibration from target families on standardized descriptors, $\hat{w}(d) = \frac{n_0}{n_1} \exp\{\hat{f}(d)\}$ with $\hat{f}$ the fitted log-odds and $n_0$, $n_1$ the class sizes, rescaled to mean one over the fitting calibration families and clipped to $[1/20, 20]$. The descriptors measure sequence similarity (maximum identity to any training protein), alignment depth ($\log(1+N_{\text{eff}})$ and log of $N_{\text{eff}}$ per position, $N_{\text{eff}}$ the effective number of sequences in the alignment) and embedding distance (to the nearest training protein, from ESM2 [28]), with log length (Appendix B). *Cross-fitting:* a third of the calibration families fits $\hat{w}$ and the rest calibrate; the target families form five folds, and the ratio for queries in fold $j$ is fitted against the target families outside fold $j$, so each fold's guarantee is conditional on its own ratio (Lemma 1).

*Computation.* We solve (3) through its dual linear program, which has one bounded variable per calibration score but only $p$ equality constraints; test families change only the right-hand side, so HiGHS [21] re-solves each from the previous basis (a few milliseconds). To obtain a value at least the supremum in (3) when $\arg\min L$ is not a single point, the test term's weight is raised from $\tau$ to $\tau + 10^{-4}$: every exact minimizer of the perturbed program has a fitted value at least the supremum, and an unbounded perturbed program gives $+\infty$ (Appendix A). Solvers are optimal only within tolerances, so the returned basis is certified optimal for the perturbed program in exact rational arithmetic (a cold re-solve, then an exact dual simplex, as fallbacks), and the fitted value is rounded up. The computed sets therefore contain those of (3) and keep every guarantee by Lemma 2 (Appendix B reports the checks).

*(iii) Support gate.* The gate returns $\{0,1\}$ for every variant of a query family (it abstains) when few calibration families resemble it, measured by a kernel effective sample size (ESS) of the support descriptors (maximum identity, alignment depth, embedding distance) below $m$, or when the ratio puts more than $\pi_{\max}$ of ProtCOVER-W's quantile mass on the query itself; its parameters are tuned on an inner split of the training families (Appendix B). It only enlarges sets, so results for ProtCOVER and its variants are without it.

*(iv) Class-conditional versions.* Every component has a Mondrian version: for each label $y$, the threshold $\hat{q}_y$ is computed from label-$y$ calibration variants only (weights $1/N_{k,y}$; families with $N_{k,y} = 0$ dropped), and $y \in C(x)$ iff $s(x,y) \le \hat{q}_y$. These target $\text{Cov}_0$ and $\text{Cov}_1$ separately and are the versions we compare.

## 5 Guarantees

All results are finite-sample and hold for any fixed score function, so they do not depend on the quality of the base model or of the predictor; proofs are in Appendix A. Each concerns the expected coverage of a random variant of a new family, not the coverage of every family. Table 1 lists what each covers.

**Proposition 1 (Family-level calibration).** *Under A1, the set* (2) *with* $\omega \equiv 1$ *satisfies* $\text{Cov}(C) \ge 1 - \alpha$.

Proposition 1 is Theorem 1 of [25] with the test point replaced by a uniformly random variant of the test family, which removes the need for exchangeability within families.

**Theorem 1 (Coverage over a class of family-level shifts).** *Let $P$ be a law of families and $\Phi$ fixed. For $f \in \mathcal{F}$ with $f(D) \ge 0$ $P$-almost surely and $0 < \mathbb{E}_P f(D) < \infty$, let $Q_f$ be the family law with $dQ_f/dP = f(D)/\mathbb{E}_P f(D)$. If $G_1, \ldots, G_K$ are i.i.d. $P$ and, independently, $G_{K+1} \sim Q_f$, the set* (3) *satisfies* $\text{Cov}(C) \ge 1 - \alpha$. *The same holds under A1 ($f \equiv 1$; the families need only be exchangeable). For fitted features such as $(1, \hat{w})$, the guarantee is conditional on the fit and relies on cross-fitting's independence (Lemma 1).*

**Corollary 1 (Any family-level shift).** *With $G_1, \ldots, G_K$ i.i.d. $P$ as in Theorem 1 and a test family with any law $Q$, $\text{Cov}(C) \ge 1 - \alpha - \inf_f d_{\text{TV}}(Q, Q_f)$, the infimum over the $f$ of Theorem 1. If $\Phi = (1, \hat{w})$ for a ratio estimate $\hat{w} \ge 0$ independent of the families, with $0 < \mathbb{E}_P \hat{w} < \infty$ and $d\hat{Q}/dP = \hat{w}(D)/\mathbb{E}_P \hat{w}(D)$, then*

$$\mathbb{E}\big[\gamma_{K+1}(C) \,\big|\, \hat{w}\big] \;\ge\; 1 - \alpha - \min\big\{d_{\text{TV}}(Q,P),\; d_{\text{TV}}(Q,\hat{Q})\big\}.$$

Here $d_{\mathrm{TV}}(Q, P)$ bounds the loss of family-level calibration alone (the same argument applied to Proposition 1) and $d_{\mathrm{TV}}(Q, \hat{Q})$ that of reweighting by $\hat{w}$ (Proposition 2), so conditioning on the estimated ratio is never worse, in these bounds, than either. Both distances are unobservable, so the corollary motivates conditioning on both but certifies nothing under an arbitrary shift. Conditioning needs no total-variation term under family exchangeability, when $\hat{w}$ is the true ratio up to scale, and for every shift whose ratio is a nonnegative function in the span of 1 and $\hat{w}$; with descriptor features (ProtCOVER-D), for every shift whose ratio is nonnegative and affine in the descriptors, without target families.

*What is new.* Table 1 places these results among related guarantees. A family's miss rate $1 - \hat{F}_k(q)$ is a bounded loss that does not increase with the threshold, and family-weighted coverage is one minus its risk, which is why pooling families is the risk-control view applied to grouped data (Section 1); (3) applies conditional risk control to families with the test family's loss imputed as one, and constant features give HCP's threshold up to ties (Appendix A). For validity as computed, the largest optimal fitted value dominates the threshold of the program with the test family's true scores (step (d) of the proof). This adapts Gibbs et al.'s Proposition 6, which imputes an upper bound for the unknown scores and assumes min-norm solutions; Blot et al. treat neither ties nor unbounded programs, and Section 4 computes the threshold exactly. Conditioning on the span of a constant and a cross-fitted ratio (Gibbs et al. note a single estimated shift as a one-element class) then gives Corollary 1. Applying Gibbs et al. to one random variant per family, the analogue of subsampling once [14], also gives family-weighted coverage over $\mathcal{F}$, because the subsampled pairs are exchangeable points, but discards all but one calibration variant of each family; we use it as a baseline. Duchi et al. [13, Cor. 7.27] regress one coverage threshold per environment on environment features, which gives a new family the conservative level $(1 - \alpha)(1 - \delta)$, without pooling its variants.

**Proposition 2 (Reweighting variant).** *Under A2, the set* (2) *with* $\omega_k = w(D_k)$, $k \le K + 1$, *satisfies* $\mathrm{Cov}(C) \ge 1 - \alpha$. *With* $\omega_k = \hat{w}(D_k)$ *for* $\hat{w} \ge 0$ *independent of the families with* $0 < \mathbb{E}_P \hat{w}(D) < \infty$ *and* $\hat{Q}$ *as in Corollary 1,* $\mathbb{E}[\gamma_{K+1}(C) \mid \hat{w}] \ge 1 - \alpha - d_{\mathrm{TV}}(Q, \hat{Q})$.

Proposition 2 lifts weighted exchangeability [41] and its estimated-weight bound [27, Thm. 3] from variants to families.

**Lemma 1 (Cross-fitting).** *For each fold* $j$, *the ratio* $\hat{w}_j$ *is a function only of the fitting calibration families and of the target families outside fold* $j$. *With independent families (as under A2) it is therefore independent of the calibration families used in* (2) *and* (3) *and of every family in fold* $j$, *and Corollary 1 and Proposition 2 hold for the queries of fold* $j$ *conditionally on* $\hat{w}_j$. *If instead all non-training families are exchangeable, as under a random split, then conditionally on the families that fit* $\hat{w}_j$ *the remaining calibration families and each family of fold* $j$ *are exchangeable, and Theorem 1 with* $f \equiv 1$ *holds conditionally on* $\hat{w}_j$.

**Proposition 3 (Class-conditional versions).** *For* $y \in \{0, 1\}$ *such that a family has* $N_y > 0$ *with positive probability under* $P$ *and under the test law, the Mondrian versions of Propositions 1 and 2, Theorem 1, Corollary 1 and Lemma 1 hold with* $\mathrm{Cov}_y$ *in place of* Cov *and with* $Q$, $P$, $Q_f$, $\hat{Q}$ *conditioned on* $N_{K+1,y} > 0$.

**Lemma 2 (Enlargement).** *If* $C'(x) \supseteq C(x)$ *for all* $x$ *almost surely, where* $C'$ *may depend on all observed data, then* $\mathrm{Cov}(C') \ge \mathrm{Cov}(C)$ *and* $\mathrm{Cov}_y(C') \ge \mathrm{Cov}_y(C)$.

The gate (iii) only replaces a set by $\{0, 1\}$, so by Lemma 2 it preserves every guarantee above, whatever its parameters; it gives no guarantee for the families it retains.

## 6 Experiments

### 6.1 Data, predictors, and families

We use the ProteinGym v1.3 clinical substitution benchmark [32]: 62,727 ClinVar [24] missense variants in 2,525 human proteins, 51.0% pathogenic, with a median of 12 labeled variants per protein (maximum 741). We take ProteinGym's precomputed zero-shot scores for five unsupervised predictors: ESM-1b (protein language model), EVE and GEMME (alignment-based), TranceptEVE (hybrid), and PoET (retrieval-augmented language model), oriented so that higher means more pathogenic; our per-protein mean AUCs

**Table 1.** Guarantees of related methods and of ProtCOVER's versions: finite-sample lower bounds of $1-\alpha$ on the target under the stated conditions, unless noted (repeated subsampling: $1-2\alpha$; multi-environment: with probability $1-\delta$); "bound": reduced by a total-variation distance that unlabeled data cannot estimate. ProtCOVER's versions hold for Cov and $\text{Cov}_y$ (Proposition 3) and, with cross-fitted ratios, conditionally on each fold's ratio (Lemma 1).

| Method | Unit | Target | Guaranteed under | Target data |
|---|---|---|---|---|
| Split, Mondrian [26,43] | variant | a new variant (of class $y$) | exchangeable variants | none |
| Weighted [41,27] | variant | a new variant | covariate shift with the true ratio; estimated: bound | unlabeled |
| GWCP [5] | variant | a new variant | shift of the proportions of known groups | group proportions |
| Conditional calibration [18,7] | variant | a new variant | every shift with ratio $f(X)$, $f \in \mathcal{F}$, $f \geq 0$ | none |
| HCP, subsampling [25,14] | family | a variant of a new family | A1 | none |
| Multi-environment [13, Cor. 7.27] | environment | a new environment, w.p. $1-\delta$ | shifts in a class of environment features | none |
| Conditional calibration, one variant per family | family | Cov | every shift with ratio $f(D)$, $f \in \mathcal{F}$, $f \geq 0$ (incl. A1) | as $\mathcal{F}$ needs |
| ProtCOVER-W (Prop. 2) | family | Cov, $\text{Cov}_y$ | A2 with the true ratio; estimated: bound | unlabeled families |
| ProtCOVER-D (Thm. 1) | family | Cov, $\text{Cov}_y$ | every shift with ratio affine in the descriptors, $\geq 0$ (incl. A1) | none |
| ProtCOVER (Thm. 1, Cor. 1) | family | Cov, $\text{Cov}_y$ | every shift with ratio $a + b\,\hat{w}(D) \geq 0$ (incl. A1); otherwise bound | unlabeled families |
| Gate (Lemma 2) | family | | never lowers coverage; nothing on the retained families | |
| Not guaranteed | coverage of any given family; the splits S2–S5, where A2 holds at best approximately and is violated in S5 | | | |

match ProteinGym's published ones (0.892–0.920) to three decimals. Supervised clinical predictors, trained on ClinVar, are excluded.

Families are MMseqs2 [39] clusters of wild-type sequences at 30% identity and 80% bidirectional coverage: 2,081 families, 1,805 of them single proteins. Three alternative definitions serve as sensitivity checks (Appendix B): 50% identity (2,275 families), single-linkage components of significant homology (1,939), and, strictest, components of *any* significant hit (1,024, the largest with 867 proteins; a stress test for leakage, not a biological family). Descriptors are computed without labels (Section 4); a family is an *orphan* if no member has a significant hit at $\geq 30\%$ identity outside the family (1,276 families).

### 6.2 Protocol

Each of ten seeds splits families into 40% training, 30% calibration and 30% test (832, 624 and 625 families). **S1** draws all three at random. **S2** keeps the S1 training families and samples test families with probability proportional to $\exp(-\beta z)$, $z$ the mean of standardized maximum identity to training and $\log(1+N_{\text{eff}})$, $\beta \in \{0.5, 1, 2\}$; it shifts the very descriptors that ProtCOVER and GWCP use, so it favors them by construction.

Two shifts are defined without those descriptors, though they move them: **S3** draws test families from orphan families (mean maximum identity to training 0.048 against 0.236 for calibration families), and **S4** holds out whole regions of protein space, $k$-means clusters ($k = 20$) of the leading principal components of family-mean ESM2 embeddings, until the test set is full; a strict variant, **S4s**, holds out only whole regions (Appendix B). **S5**, a stress test that violates A2 on purpose, samples test families as S2 does, with weight $\exp(-z)$, $z$ a family's standardized within-family AUC averaged over predictors, so that predictors err more on them, a shift no label-free descriptor encodes. S4, S4s and S5 were declared before their first run.

The base model, fitted on training families only, is a logistic regression on the raw score, so that each variant is scored on its own, as in deployment; its test AUC is 0.872–0.916. A transductive variant adds the score standardized within its protein and the family's number of scored variants as a descriptor; both are label-free but need the protein's whole variant panel, and we report it as a robustness check. Baselines are variant-level split conformal (B1) and its class-conditional (Mondrian) version (B2); HCP pooling and Dunn et al.'s single and repeated subsampling (B3) [25,14]; GWCP (B4) [5] with nine descriptor bins and target proportions from the test families; and support-only and random abstention at the gate's abstention rate (B5). ProtCOVER uses 416 of the calibration families in (2)–(3) and the rest to fit $\hat{w}$; ProtCOVER-D uses all calibration families, and the gate has a pre-declared 10% abstention budget. Test families are used only for evaluation, and only their label-free descriptors enter $\hat{w}$. Appendix B lists every setting.

*Comparison criterion.* Because family-level calibration can err in either class (for some predictors it over-covers pathogenic and under-covers benign variants), we compare class-conditional methods by the *worse-class shortfall* $\max\{0, 1 - \alpha - \min(\mathrm{Cov}_0, \mathrm{Cov}_1)\}$, per predictor and setting, and treat differences below 0.01 as no difference, the criterion that we pre-registered for the external evaluation. The margin is statistical, not clinical: it lies between the median bootstrap half-width of a cell's class-wise coverage (0.005) and the median seed-to-seed s.d. (0.013). Intervals are 95% family-bootstrap intervals that resample test families within each seed (paired across methods); they are conditional on the splits, and since all seeds resplit one pool of families, inference is limited to this benchmark. Unless stated otherwise $\alpha = 0.1$.

*External evaluations.* Two evaluations calibrate on benchmark variants and test on ClinVar variants that played no part in development. Each follows a protocol pre-registered in the public repository before any method was run on its data, with amendments and post-hoc decisions explicitly documented (Appendix B). Both use the ESM-1b scores of Brandes et al. [9] and, as primary analysis, the raw-score base model (Appendix B). The *temporal* evaluation tests, in the benchmark's splits, on 14,510 variants in 1,731 benchmark proteins that entered ClinVar after its snapshot: new variants and labels, not new proteins. The *new-protein* evaluation tests on 101,197 variants (9.4% pathogenic) in 12,225 human proteins outside the benchmark, clustered as for the benchmark into 8,579 test families (primary population) of which 4,448 have no detectable benchmark homolog (secondary); training families are each seed's S1 draw and every other benchmark family calibrates. New proteins have no alignments, so the descriptors are alignment-free. A second predictor, ESM2 with 150M parameters scored by us on both sides [28,31], was pre-registered before any result.

### 6.3 Results

*Variant-level calibration misses pathogenic variants per family.* In S1 (Appendix Fig. 1), variant-level split conformal has variant-weighted coverage 0.883–0.899 and family-weighted coverage 0.871–0.903; the failure is in the pathogenic class. Mondrian split conformal (B2) attains variant-weighted pathogenic coverage 0.895–0.904 for four predictors yet covers only 0.823–0.873 of pathogenic variants per family for all five. The 5% of test families with the most pathogenic variants hold 50% of them, and their pathogenic variants are easier: under B2, families with fewer than five pathogenic variants have pathogenic coverage 0.781–0.854 against 0.897–0.906 for families with at least twenty. For TranceptEVE even the variant-level target is missed (0.864; a family-level split does not make variants exchangeable), and its benign coverage per family is 0.802. Family-level class-conditional calibration attains 0.900–0.904 (pathogenic) and 0.897–0.902 (benign), as Proposition 3 guarantees in expectation, and ProtCOVER 0.902–0.904 and 0.898–0.908 with mean set sizes 1.171–1.405 (HCP 1.162–1.425): without shift, conditioning on the ratio costs little.

*Under descriptor shift ProtCOVER has the smallest shortfall; on the other shifts it matches group weighting.* We read S2, which is defined by the descriptors that ProtCOVER and GWCP use, separately from S3 and

**Table 2.** Worse-class shortfall below the 0.90 target, $\max\{0, 0.9 - \min(\mathrm{Cov}_0, \mathrm{Cov}_1)\}$, of class-conditional family-level methods at $\alpha = 0.1$. Benchmark and temporal rows: mean (maximum) over the row's predictor-setting cells, S2 pooling $\beta \in \{0.5, 1, 2\}$; a maximum of 0.000 means the target is met in every cell. New-protein rows: the single result of that predictor and population (the row on benchmark lengths is exploratory). S5 violates A2. Last two rows: mean set size over the benchmark's shifted cells, at $\alpha = 0.1$ and at pathogenic coverage 0.90 (interpolated).

| | | | ProtCOVER | | |
|---|---|---|---|---|---|
| Evaluation | HCP | GWCP | -W | -D | **default** |
| *Benchmark, five predictors (development data)* | | | | | |
| S2, descriptor shift | 0.029 (0.080) | 0.010 (0.024) | 0.001 (0.007) | 0.005 (0.016) | 0.000 (0.004) |
| S3, orphan families | 0.008 (0.017) | 0.003 (0.008) | 0.003 (0.007) | 0.001 (0.003) | 0.003 (0.008) |
| S4, held-out regions | 0.003 (0.013) | 0.003 (0.006) | 0.007 (0.013) | 0.004 (0.011) | 0.000 (0.000) |
| S4s, whole regions only | 0.005 (0.013) | 0.004 (0.009) | 0.002 (0.005) | 0.000 (0.001) | 0.000 (0.000) |
| S5, label-based shift | 0.122 (0.139) | 0.120 (0.138) | 0.118 (0.134) | 0.114 (0.130) | 0.116 (0.129) |
| *External ClinVar variants after the snapshot, ESM-1b, S2–S4 (pre-registered)* | | | | | |
| Raw-score base model (primary) | 0.004 (0.020) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) |
| Transductive base model | 0.005 (0.023) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) |
| *Proteins outside the benchmark, alignment-free (pre-registered; all new families primary)* | | | | | |
| ESM-1b, all new families | 0.036 | 0.034 | 0.032 | 0.025 | 0.027 |
| ESM-1b, no benchmark homolog | 0.083 | 0.079 | 0.077 | 0.048 | 0.062 |
| ESM-1b, benchmark lengths only | 0.021 | 0.019 | 0.017 | 0.011 | 0.012 |
| ESM2, all new families | 0.022 | 0.021 | 0.019 | 0.015 | 0.014 |
| ESM2, no benchmark homolog | 0.035 | 0.033 | 0.030 | 0.035 | 0.026 |
| *Mean set size, benchmark S2–S4* | | | | | |
| At $\alpha = 0.1$ | 1.228 | 1.241 | 1.262 | 1.252 | 1.294 |
| At matched $\mathrm{Cov}_1 = 0.90$ | 1.266 | 1.259 | 1.263 | 1.253 | 1.268 |

S4, which are not (Table 2). In S2, ProtCOVER has the smallest mean worse-class shortfall (0.000; GWCP 0.010, HCP 0.029), smaller than GWCP's by more than the 0.01 margin in 4 of the 15 S2 cells (all at $\beta = 2$, where its sets are larger) and than HCP's in 13. On S3 and S4 it is 0.002, against 0.003 for GWCP and 0.006 for HCP; paired intervals put ProtCOVER's worse-class coverage above GWCP's in 8 of 10 cells and below in 1 (differences $-0.002$ to $+0.012$), but the shortfalls are within the margin in every cell. Over all 25 shifted cells, ProtCOVER's worst cell is 0.892 (HCP 0.820, GWCP 0.876), and its shortfall never exceeds a baseline's by more than the margin. On S4, reweighting alone (ProtCOVER-W) falls short by 0.007 (GWCP 0.003), while ProtCOVER, which keeps the constant feature, meets the target in every S4 cell. The ordering holds at $\alpha = 0.05$ and 0.2 (mean shortfall 0.000 and 0.002 for ProtCOVER, 0.005 and 0.011 for GWCP; Appendix Tables 12 and 13), at $\alpha = 0.05$ with built-in abstention on up to 2.8% of test families.

*The price is a more conservative operating point, not less efficient sets.* ProtCOVER adapts how conservative it is to the shift: its mean worse-class coverage is 0.903 in S1 (HCP 0.900) but 0.906 over the shifted cells (GWCP 0.893, HCP 0.881; Appendix Fig. 2), so its sets under shift are larger; at matched pathogenic coverage the family-level methods need about the same set size (last two rows of Table 2; Appendix Fig. 3).

*A label-based shift defeats every method, and the weights do not see it.* In S5, every family-level method falls to a worse-class coverage of 0.761–0.825 (Table 2), while the effective sample size $(\sum_k \hat{w}_k)^2 / \sum_k \hat{w}_k^2$ of the calibration weights is 0.98 of the number of families, as in S1 (0.99; S2 at $\beta = 2$: 0.26): the weights neither correct nor flag such a shift, and a weight diagnostic can show that a shift is visible, never that none exists.

*Robustness, and method against information.* The ranking under shift is unchanged with the transductive base model and, on S4s and under the three alternative family definitions, ProtCOVER's mean shortfall is

again at most GWCP's and HCP's (Table 2; Appendix Table 5). Without alignment depth, the descriptor that drives the S2 correction, the shift-aware methods fall further short, but ProtCOVER's shortfall is still smaller than GWCP's by more than the margin in 4 cells and larger in none. Conditional calibration on one random variant per family, which already attains family-weighted coverage, falls short by 0.002 over the shifted cells (ProtCOVER 0.001), with the same set size at matched coverage but coverage that varies more across seeds; richer feature classes are more conservative, and other ratio-fit shares change ProtCOVER's shortfall by less than the margin (Appendix D). With ProtCOVER's information, GWCP on ninths of $\hat{w}$ falls short by 0.006 (its own bins: 0.007); with GWCP's, conditional calibration on its bins falls short by 0.000 but abstains on up to 13.4% of a cell's test families (ProtCOVER 0.3%; Appendix Table 6). The gain over GWCP comes mainly from conditioning, not the ratio's information; conditioning on the ratio, not nine bins, keeps abstention rare.

*New variants of benchmark proteins.* The pre-registered hypotheses of the temporal evaluation were supported (Appendix B; H4's rule was fixed after the run; Mondrian split covers 0.873 [0.864, 0.883] of pathogenic variants per family in S1), but ESM-1b separates these variants better (per-protein AUC 0.930, against 0.899 on their benchmark variants), and the data do not discriminate between the shift-aware methods: GWCP, ProtCOVER and its variants meet the target in every shifted setting; only HCP falls short (0.004).

*On proteins outside the benchmark, every method under-covers pathogenic variants, the conditional versions the least.* The pre-registered hypotheses were that Mondrian split covers less than 0.90 of pathogenic variants per family (U1), that ProtCOVER's worse class covers at least 0.89 (U2), and that ProtCOVER's shortfall exceeds neither HCP's nor GWCP's by 0.01 or more (U3) and is smaller than GWCP's by more than 0.01 (U4). On the new proteins, Mondrian split under-covers pathogenic variants per family (0.778 [0.771, 0.786]; U1 supported; `docs/unseen_results.md`). Every family-level method falls short of the target in the pathogenic class (Table 2) while over-covering benign variants (0.965 to 0.967); ProtCOVER's worse class covers 0.873, so U2 is not supported. ProtCOVER's shortfall (0.027) exceeds neither HCP's (0.036) nor GWCP's (0.034) by the margin (U3 supported), but is not smaller than GWCP's by more than it (U4 not supported); without detectable homologs the gap to GWCP exceeds it (0.062 against 0.079). ProtCOVER-D, outside the hypotheses, falls short least (0.025; without homologs 0.048). At matched coverage ProtCOVER's sets are no larger (1.366, GWCP 1.372). The density ratio barely departs from one (effective sample size 0.96 of the calibration families), while ESM-1b separates the new proteins' variants worse (per-protein AUC 0.873 against 0.892). The 223 families with a protein longer than any benchmark protein carry much of the shortfall (without them, exploratory, ProtCOVER's is 0.012), and neither the ratio nor the gate flags them (ProtCOVER abstains on 0.0% of families). The ratio includes log length, but that does not guarantee detecting proteins outside the calibration range: benchmark proteins are longer on average, the fitted length coefficient is negative in every seed, and logistic extrapolation places these families near the bottom of $\hat{w}$ (median rank 1.5% among new families, mean $\hat{w}$ 0.56 against 1.10), none of them clipped. The gate's descriptors do not include length: 14.1% of these families lie in the lowest decile of its ESS, and, tuned on the benchmark, it abstains on no new family. These are plausible explanations from exploratory diagnostics, not proven causes. With ESM2 (per-protein AUC 0.795 on the new proteins, 0.808 on the benchmark), the four verdicts are the same (Mondrian split 0.850) and the conditional versions again fall short least (ProtCOVER 0.014, ProtCOVER-D 0.015; HCP 0.022, GWCP 0.021), but without homologs ProtCOVER's gap to GWCP is within the margin (0.026 against 0.033) and ProtCOVER-D falls short as much as HCP (0.035).

*The support gate preserves coverage but does not improve selective risk.* We keep the gate as an informative negative experiment, not a benefit of ProtCOVER. Abstaining first, on ProtCOVER-W's sets, on the families with the lowest ESS lowered the area under the risk-coverage curve (AURC) relative to random abstention in the majority of seeds in 13 of 30 cells and raised it in 14 (Appendix Fig. 4); ordering by maximum identity alone did better (lower in 23 cells, higher in 6), with small mean differences from random ($-0.011$ to $+0.017$). Support descriptors locate the families whose coverage needs correcting, which conditional calibration already does; once coverage is controlled they predict little of where the predictor errs.

## 7 Limitations

*What the guarantees assume.* Coverage of at least $1 - \alpha$ needs exchangeable families (A1), or a family-level shift whose density ratio is a nonnegative function in the span of the features, with an estimated ratio fitted on families independent of those it calibrates (Lemma 1); the class-conditional versions hold for families that contain the class. Every guarantee is marginal over families: ProtCOVER certifies no particular protein, and the coverage of a small family is dominated by binomial noise. Under any other shift, Corollary 1 bounds the loss by total-variation distances that unlabeled data cannot estimate; when a family's scores and labels given its descriptors differ between calibration and deployment (A2 fails, as in S5 and apparently on the new proteins), every method under-covers and the ratio gives no warning. Including log length does not guarantee flagging test families outside the calibration families' range, since logistic extrapolation can rank them as typical of calibration, as with the longest new proteins, and the gate's descriptors omit length.

*What is only measured.* Every comparison with baselines is empirical. Beyond the 0.01 margin, ProtCOVER's advantage over group-weighted conformal prediction appears mainly under strong descriptor shift (S2 at $\beta = 2$), on one benchmark whose ten seeds resplit one pool of families, and, with ESM-1b but not ESM2, among new families without a detectable benchmark homolog (a secondary population); elsewhere it matches group weighting within the margin, and on proteins outside the benchmark no method reaches the target. ProtCOVER buys its smaller shortfall with larger sets under shift, and conditional calibration on one variant per family is within the margin of ProtCOVER in all but 1 of the 25 shifted cells.

*Scope and deployment.* ProtCOVER is appropriate when a batch of unlabeled target families is available (a reference panel such as the unlabeled proteome could serve a single query, but the guarantee would then concern the panel's shift); ProtCOVER-D can be used without one. Neither guarantees target coverage when predictor accuracy changes in ways the descriptors cannot capture. ProtCOVER-D's shortfall was larger than ProtCOVER's under descriptor shift but smaller on orphan families and, with ESM-1b, on the new proteins, and ProtCOVER's default was chosen on the benchmark, which is development data. ProtCOVER outputs prediction sets, not ACMG/AMP evidence strengths; mapping sets to PP3/BP4 strengths is future work.

## 8 Conclusion

Calibrating a variant effect predictor variant by variant lets well-studied families set its error rate: on ProteinGym, Mondrian split conformal covers only 0.823–0.873 of pathogenic variants per family for all five predictors, without any shift. Treating families as the exchangeable unit removes this failure, and conditioning a family-weighted quantile regression on a cross-fitted density ratio guarantees coverage under family exchangeability and when the estimated ratio is correct. The empirical gain is in coverage and appears mainly under descriptor shift (S2); on orphan families and held-out embedding regions ProtCOVER matches group weighting. On proteins outside the benchmark, a pre-registered test with two predictors found variant-level calibration failing again and every method short of the target where the predictors were less accurate and the longest proteins lacked calibration support, neither of which the label-free ratio or the gate flags; conditioning narrowed the shortfall, within the margin of group weighting, without closing it. Descriptors that track a predictor's accuracy without labels, and calibration support for proteins unlike any calibration protein, remain open.

*Code availability.* Code and scripts that regenerate every number, table and figure from the public ProteinGym release: `https://github.com/wael-fahmy/protcover` (MIT license).

## References


1. Angelopoulos, A.N., Bates, S.: Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning **16**(4), 494–591 (2023). `https://doi.org/10.1561/2200000101`
2. Angelopoulos, A.N., Bates, S., Fisch, A., Lei, L., Schuster, T.: Conformal risk control. In: International Conference on Learning Representations (2024), arXiv:2208.02814

3. Bairaktari, K., Wu, J., Wu, Z.S.: Kandinsky conformal prediction: Beyond class- and covariate-conditional coverage. arXiv preprint arXiv:2502.17264 (2025)
4. Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J.: Conformal prediction beyond exchangeability. The Annals of Statistics **51**(2), 816–845 (2023). https://doi.org/10.1214/23-AOS2276
5. Bhattacharyya, A., Barber, R.F.: Group-weighted conformal prediction. arXiv preprint arXiv:2401.17452 (2024)
6. Bhattacharyya, A., Ding, T., Barber, R.F.: Conformal prediction with macro-coverage guarantees. arXiv preprint arXiv:2606.28598 (2026)
7. Blot, V., Angelopoulos, A.N., Jordan, M.I., Brunel, N.J.B.: Automatically adaptive conformal risk control. In: Proceedings of the 28th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 258 (2025), arXiv:2406.17819
8. Boger, R.S., Chithrananda, S., Angelopoulos, A.N., Yoon, P.H., Jordan, M.I., Doudna, J.A.: Functional protein mining with conformal guarantees. Nature Communications **16**, 85 (2025). https://doi.org/10.1038/s41467-024-55676-y
9. Brandes, N., Goldman, G., Wang, C.H., Ye, C.J., Ntranos, V.: Genome-wide prediction of disease variant effects with a deep protein language model. Nature Genetics **55**, 1512–1522 (2023). https://doi.org/10.1038/s41588-023-01465-0
10. Cauchois, M., Gupta, S., Ali, A., Duchi, J.C.: Robust validation: Confident predictions even when distributions shift. Journal of the American Statistical Association (2024). https://doi.org/10.1080/01621459.2023.2298037
11. Chen, Y., Fayer, S., Jain, S., Benazouz, M., et al.: Gene- and domain-aware calibration increases the clinical utility of variant effect predictors. bioRxiv (2026). https://doi.org/10.64898/2026.02.17.706269
12. Ding, T., Angelopoulos, A.N., Bates, S., Jordan, M.I., Tibshirani, R.J.: Class-conditional conformal prediction with many classes. In: Advances in Neural Information Processing Systems. vol. 36 (2023), arXiv:2306.09335
13. Duchi, J.C., Gupta, S., Jiang, K., Sur, P.: Predictive inference in multi-environment scenarios. arXiv preprint arXiv:2403.16336 (2024)
14. Dunn, R., Wasserman, L., Ramdas, A.: Distribution-free prediction sets for two-layer hierarchical models. Journal of the American Statistical Association **118**(544), 2491–2502 (2023). https://doi.org/10.1080/01621459.2022.2060112
15. Fannjiang, C., Bates, S., Angelopoulos, A.N., Listgarten, J., Jordan, M.I.: Conformal prediction under feedback covariate shift for biomolecular design. Proceedings of the National Academy of Sciences **119**(43), e2204569119 (2022). https://doi.org/10.1073/pnas.2204569119
16. Frazer, J., Notin, P., Dias, M., Gomez, A., Min, J.K., Brock, K., Gal, Y., Marks, D.S.: Disease variant prediction with deep generative models of evolutionary data. Nature **599**, 91–95 (2021). https://doi.org/10.1038/s41586-021-04043-8
17. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems. vol. 30 (2017), arXiv:1705.08500
18. Gibbs, I., Cherian, J.J., Candès, E.J.: Conformal prediction with conditional guarantees. Journal of the Royal Statistical Society Series B: Statistical Methodology **87**(4), 1100–1126 (2025). https://doi.org/10.1093/jrsssb/qkaf008
19. Groth, P.M., Kerrn, M.H., Olsen, L., Salomon, J., Boomsma, W.: Kermut: Composite kernel regression for protein variant effects. In: Advances in Neural Information Processing Systems. vol. 37 (2024), arXiv:2407.00002
20. Guan, L.: Localized conformal prediction: A generalized inference framework for conformal prediction. Biometrika **110**(1), 33–50 (2023). https://doi.org/10.1093/biomet/asac040
21. Huangfu, Q., Hall, J.A.J.: Parallelizing the dual revised simplex method. Mathematical Programming Computation **10**(1), 119–142 (2018). https://doi.org/10.1007/s12532-017-0130-5
22. Jung, C., Noarov, G., Ramalingam, R., Roth, A.: Batch multivalid conformal prediction. arXiv preprint arXiv:2209.15145 (2022)
23. Laine, E., Karami, Y., Carbone, A.: GEMME: A simple and fast global epistatic model predicting mutational effects. Molecular Biology and Evolution **36**(11), 2604–2619 (2019). https://doi.org/10.1093/molbev/msz179

24. Landrum, M.J., et al.: Clinvar: Improving access to variant interpretations and supporting evidence. Nucleic Acids Research **46**(D1), D1062–D1067 (2018). https://doi.org/10.1093/nar/gkx1153
25. Lee, Y., Barber, R.F., Willett, R.: Distribution-free inference with hierarchical data. ACM/IMS Journal of Data Science (2026). https://doi.org/10.1145/3786352
26. Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R.J., Wasserman, L.: Distribution-free predictive inference for regression. Journal of the American Statistical Association **113**(523), 1094–1111 (2018). https://doi.org/10.1080/01621459.2017.1307116
27. Lei, L., Candès, E.J.: Conformal inference of counterfactuals and individual treatment effects. Journal of the Royal Statistical Society Series B: Statistical Methodology **83**(5), 911–938 (2021). https://doi.org/10.1111/rssb.12445
28. Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., Shmueli, Y., dos Santos Costa, A., Fazel-Zarandi, M., Sercu, T., Candido, S., Rives, A.: Evolutionary-scale prediction of atomic-level protein structure with a language model. Science **379**(6637), 1123–1130 (2023). https://doi.org/10.1126/science.ade2574
29. Liu, Y., Levis, A.W., Normand, S.L., Han, L.: Multi-source conformal inference under distribution shift. In: Proceedings of the 41st International Conference on Machine Learning. PMLR, vol. 235 (2024), arXiv:2405.09331
30. Mallick, S., Tchetgen Tchetgen, E., Dobriban, E., Lee, Y.: Generalized hierarchical conformal prediction. arXiv preprint arXiv:2608.15500 (2026)
31. Meier, J., Rao, R., Verkuil, R., Liu, J., Sercu, T., Rives, A.: Language models enable zero-shot prediction of the effects of mutations on protein function. In: Advances in Neural Information Processing Systems. vol. 34 (2021). https://doi.org/10.1101/2021.07.09.450648
32. Notin, P., Kollasch, A.W., Ritter, D., van Niekerk, L., Paul, S., Spinner, H., Rollins, N., Shaw, A., Orenbuch, R., Weitzman, R., Frazer, J., Dias, M., Franceschi, D., Gal, Y., Marks, D.S.: Proteingym: Large-scale benchmarks for protein design and fitness prediction. bioRxiv (2023). https://doi.org/10.1101/2023.12.07.570727, also in Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track
33. Notin, P., Van Niekerk, L., Kollasch, A.W., Ritter, D., Gal, Y., Marks, D.S.: TranceptEVE: Combining family-specific and family-agnostic models of protein sequences for improved fitness prediction. bioRxiv (2022). https://doi.org/10.1101/2022.12.07.519495
34. Passi, A., Schneidman-Duhovny, D.: Calibrated variant effect prediction at the residue level using conditional score distributions. bioRxiv (2025). https://doi.org/10.1101/2025.11.24.690189
35. Pejaver, V., et al.: Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. The American Journal of Human Genetics **109**(12), 2163–2177 (2022). https://doi.org/10.1016/j.ajhg.2022.10.013
36. Podkopaev, A., Ramdas, A.: Distribution-free uncertainty quantification for classification under label shift. In: Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence. PMLR, vol. 161 (2021), arXiv:2103.03323
37. Richards, S., et al.: Standards and guidelines for the interpretation of sequence variants. Genetics in Medicine **17**(5), 405–424 (2015). https://doi.org/10.1038/gim.2015.30
38. Rives, A., Meier, J., Sercu, T., Goyal, S., Lin, Z., Liu, J., Guo, D., Ott, M., Zitnick, C.L., Ma, J., Fergus, R.: Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. Proceedings of the National Academy of Sciences **118**(15), e2016239118 (2021). https://doi.org/10.1073/pnas.2016239118
39. Steinegger, M., Söding, J.: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology **35**, 1026–1028 (2017). https://doi.org/10.1038/nbt.3988
40. Tejura, M., Fayer, S., McEwen, A.E., Flynn, J., et al.: Calibration of variant effect predictors on genome-wide data masks heterogeneous performance across genes. The American Journal of Human Genetics (2024). https://doi.org/10.1016/j.ajhg.2024.07.018
41. Tibshirani, R.J., Barber, R.F., Candès, E.J., Ramdas, A.: Conformal prediction under covariate shift. In: Advances in Neural Information Processing Systems. vol. 32 (2019), arXiv:1904.06019
42. Truong, Jr., T.F., Bepler, T.: PoET: A generative model of protein families as sequences-of-sequences. In: Advances in Neural Information Processing Systems. vol. 36 (2023), arXiv:2306.06156

43. Vovk, V.: Conditional validity of inductive conformal predictors. In: Proceedings of the Asian Conference on Machine Learning. PMLR, vol. 25, pp. 475–490 (2012), arXiv:1209.2673
44. Vovk, V., Gammerman, A., Shafer, G.: Algorithmic Learning in a Random World. Springer (2005). https://doi.org/10.1007/b106715

# A Proofs

Throughout, $F_k = N_k^{-1} \sum_{i=1}^{N_k} \delta_{S_{k,i}}$ is the score distribution of family $k$ (for the test family it involves unobserved labels and is used only in the analysis), and $\gamma_{K+1}(C) = F_{K+1}((-\infty, \hat{q}])$ because a test variant is covered iff its score is at most $\hat{q}$.

*Comparison step for* (2). Fix weights $p_1, \dots, p_{K+1}$ and define the oracle measure $\mu = \sum_{k=1}^{K+1} p_k F_k$ and $q = \mathrm{Quant}_{1-\alpha}(\mu)$. The computable $\hat{\mu}$ moves the mass $p_{K+1}$ from $F_{K+1}$ to $+\infty$, so $\hat{\mu}((-\infty, u]) \le \mu((-\infty, u])$ for every $u$, hence $\hat{q} \ge q$ and $\gamma_{K+1}(C) \ge F_{K+1}((-\infty, q])$. Also $\mu((-\infty, q]) \ge 1 - \alpha$ by right-continuity.

*Proof (Proposition 1).* With $\omega \equiv 1$, $p_k = 1/(K+1)$ and $q$ is a symmetric function of $(G_1, \dots, G_{K+1})$. By exchangeability, $\mathbb{E}[F_{K+1}((-\infty, q])] = \mathbb{E}[(K+1)^{-1} \sum_k F_k((-\infty, q])] = \mathbb{E}[\mu((-\infty, q])] \ge 1 - \alpha$, and the comparison step gives $\mathrm{Cov}(C) \ge 1 - \alpha$.

*Proof (Theorem 1).* Let $L^o(b) = \sum_{k=1}^{K+1} N_k^{-1} \sum_i \rho_\tau(S_{k,i} - \Phi(D_k)^\top b)$ be the oracle objective, in which the test family enters with its true scores. It is convex, piecewise linear and nonnegative, so its minimizers form a nonempty closed convex polyhedron; let $\tilde{b}$ be the one of minimum Euclidean norm (a measurable function of the data, for example as the limit of the ridge-penalized minimizers as the penalty vanishes; only its symmetry is used), $\tilde{g} = \Phi^\top \tilde{b}$, and $\tilde{\gamma}_k = N_k^{-1} \sum_i \mathbf{1}\{S_{k,i} \le \tilde{g}(D_k)\}$.

*(a) First-order condition.* Let $f = \Phi^\top b_f$ with $f(D) \ge 0$ $P$-a.s.; then $f(D_k) \ge 0$ for every $k$ almost surely, which is all the argument uses. The right derivative of $t \mapsto \rho_\tau(S - \tilde{g}(D) - t f(D))$ at $t = 0$ is $f(D)(\mathbf{1}\{S \le \tilde{g}(D)\} - \tau)$ (if $S = \tilde{g}(D)$, the residual is $-t f(D) \le 0$ for $t > 0$). The right derivative of $t \mapsto L^o(\tilde{b} + t b_f)$ at 0 is therefore $\sum_{k=1}^{K+1} f(D_k)(\tilde{\gamma}_k - \tau)$, which is nonnegative at a minimizer.

*(b) Symmetry.* $L^o$, and hence $\tilde{b}$, is invariant to permutations of the families, so the pairs $(f(D_k), \tilde{\gamma}_k)$, $k \le K+1$, are exchangeable under $P^{K+1}$ and $\mathbb{E}_{P^{K+1}}[f(D_{K+1})(\tilde{\gamma}_{K+1} - \tau)] = (K+1)^{-1} \mathbb{E} \sum_k f(D_k)(\tilde{\gamma}_k - \tau) \ge 0$; the expectation exists because $\tilde{\gamma} \in [0, 1]$ and $\mathbb{E}_P f < \infty$.

*(c) Change of measure.* For any bounded measurable $H$ of the families, $\mathbb{E}_{P^{K+1}}[f(D_{K+1}) H] = \mathbb{E}_P f(D) \cdot \mathbb{E}_{P^K \times Q_f}[H]$. With $H = \tilde{\gamma}_{K+1} - \tau$, (b) gives $\mathbb{E}_{P^K \times Q_f}[\tilde{\gamma}_{K+1}] \ge \tau$.

*(d) The computable threshold dominates the oracle.* We show $\hat{q} \ge \tilde{g}(D_{K+1})$, which gives $\gamma_{K+1}(C) \ge \tilde{\gamma}_{K+1}$ and, with (c), the theorem. The argument follows Proposition 6 of [18] (imputing an upper bound $M$), with the supremum over the solution set in place of a min-norm solution, and $M \to \infty$. Write $\phi_t = \Phi(D_{K+1})$. If $L$ in (3) is unbounded below, $\hat{q} = +\infty$. Otherwise $A = \arg\min L$ is nonempty (a polyhedral function bounded below attains its minimum). Pick $M$ larger than every test score and than $\inf_{b \in A} \phi_t^\top b$, and let $L^M$ be $L^o$ with every test score replaced by $M$. Since $\rho_\tau(u) \ge \tau u$ with equality for $u \ge 0$, $L^M(b) = L(b) + \tau \phi_t^\top b + \rho_\tau(M - \phi_t^\top b) \ge L(b) + \tau M$, with equality when $\phi_t^\top b \le M$, which holds for some $b \in A$; hence $\min L^M = \min L + \tau M$ and $\arg\min L^M \subseteq A$. Let $b' \in \arg\min L^M$, $a = \tilde{g}(D_{K+1})$ and $a' = \phi_t^\top b'$. Adding $L^o(\tilde{b}) \le L^o(b')$ and $L^M(b') \le L^M(\tilde{b})$, the calibration terms cancel and $\sum_i [h_i(a) - h_i(a')] \le 0$, where $h_i(c) = \rho_\tau(S_{K+1,i} - c) - \rho_\tau(M - c)$ is nondecreasing in $c$ because $S_{K+1,i} \le M$ and $\rho_\tau$ is convex. If $a' < a$, then $h_i(a') \le h_i(a)$ for every $i$, so both optimality inequalities hold with equality and $\tilde{b} \in \arg\min L^M \subseteq A$. In either case $\hat{q} = \sup_{b \in A} \phi_t^\top b \ge a$.

The case $f \equiv 1$ gives the statement under A1 (for exchangeable rather than i.i.d. families, (b) applies directly with $f \equiv 1$).

*Computing the supremum.* Let $L'$ be $L$ with the test term's weight raised from $\tau$ to $\tau' > \tau$, so $L'(b) = L(b) - (\tau' - \tau)\phi_t^\top b$. If $b \in \arg\min L$ and $b' \in \arg\min L'$, adding $L(b) \le L(b')$ and $L'(b') \le L'(b)$ gives $(\tau' - \tau)\, \phi_t^\top (b' - b) \ge 0$, so every minimizer of $L'$ has $\phi_t^\top b' \ge \hat{q}$. If $L$ is unbounded below along a direction $d$, its recession slope along $d$ is negative, which forces $\phi_t^\top d > 0$, so $L'$ is unbounded along $d$ as well. If $L$ is bounded but $\hat{q} = +\infty$, then $A$ contains points with $\phi_t^\top b \to \infty$ and $L'$ is unbounded below on $A$. In every case every exact minimizer of $L'$ with $\tau' = \tau + 10^{-4}$ has fitted value at least $\hat{q}$, and an unbounded $L'$ gives $+\infty$; these fitted values all equal $\hat{q}$ when $\tau' - \tau$ is small enough for the data (the optimal value is concave and piecewise linear in $\tau'$, with finitely many breakpoints, and its right derivative at $\tau$ is $-\hat{q}$), and is otherwise conservative. A floating-point solver finds a minimizer of $L'$ only up to its tolerances, and near a tie of $L$ it can return the other end of the tie, below $\hat{q}$, so every computed threshold is certified. For the basis that

the solver returns, $b$ solves the $p$ interpolation equations of its basic scores exactly; every other score's dual variable is set to the bound that the sign of its exact residual dictates; and $b$ minimizes $L'$ if the basic dual variables, which then solve a $p \times p$ system, lie within their bounds (a sufficient condition: with scores on the fitted surface it can fail for an optimal $b$, a case the exact simplex below resolves). This check is done in exact rational arithmetic, with $\tau = 1 - \alpha$, $\tau'$ and the weights $1/N_k$ exact (floating-point evaluation with rigorous error bounds decides almost every case), and the certified fitted value is rounded up. A basis can pass the solver's tolerances and fail the check: a score whose residual is below those tolerances may belong at its other bound. If neither the returned basis nor that of a cold re-solve with tight tolerances passes, a bounded dual simplex in exact rational arithmetic takes over, started from the solver's basis with every nonbasic dual variable at the bound its exact residual sign dictates (a dual-feasible start; bound-flipping ratio test, Bland's smallest-index rule on degenerate steps). It ends with an exactly optimal basis, whose optimality it checks again, or with an exact certificate that $L'$ is unbounded; the solver's unboundedness verdicts are checked the same way. Every computed threshold is therefore the exact optimal fitted value of $L'$, rounded up, except that a pivot cap, never reached, would set it to $+\infty$. By Lemma 2 the computed sets keep every guarantee for the stored double-precision scores and features. Standardizing the descriptors (ProtCOVER-D) is an invertible affine reparametrization of $b$ that does not change $\mathcal{F}$, so fitted values, and $\hat{q}$, are unchanged. With $\Phi \equiv 1$, (3) gives the HCP threshold (2), except that when the pooled distribution $\hat{\mu}$ reaches $1 - \alpha$ exactly at that threshold it takes the next pooled calibration score, or $+\infty$ if there is none.

*Proof (Corollary 1).* Let $\phi(g) \in [0, 1]$ be the expected per-family coverage of a test family $g$ over $G_1, \ldots, G_K \sim P$. For $f \in \mathcal{F}$, $f \geq 0$, Theorem 1 gives $\mathbb{E}_{Q_f} \phi \geq 1 - \alpha$, and $|\mathbb{E}_Q \phi - \mathbb{E}_{Q_f} \phi| \leq d_{\mathrm{TV}}(Q, Q_f)$. For $\Phi = (1, \hat{w})$, condition on $\hat{w}$: $f \equiv 1$ and $f = \hat{w}$ are nonnegative members of $\mathcal{F}$ with $Q_1 = P$ and $Q_{\hat{w}} = \hat{Q}$.

*Proof (Proposition 2).* Under A2 the joint law of $(G_1, \ldots, G_{K+1})$ has density $\prod_{k \leq K} \pi(g_k) \cdot w(d_{K+1}) \pi(g_{K+1})$ with respect to a product base measure, where $\pi$ is a density of $P$, so the sequence is weighted exchangeable with weight functions $1, \ldots, 1, w$ [41, Def. 1, Lemma 2]. Let $E$ be the event that the unordered collection $\{G_1, \ldots, G_{K+1}\}$ equals $\{g_1, \ldots, g_{K+1}\}$. As in the proof of Lemma 3 of [41], $\mathbb{P}(G_{K+1} = g_k \mid E) = w(d_k) / \sum_j w(d_j) = p_k$ (summing over coincident values). The $p_k$, $\mu$ and $q$ are functions of $E$, so $\mathbb{E}[F_{K+1}((-\infty, q]) \mid E] = \mu((-\infty, q]) \geq 1 - \alpha$, and the comparison step gives the first claim. For $\hat{w}$ independent of the families, the procedure is invariant to rescaling the weights, so conditionally on $\hat{w}$ it is the procedure of the first claim for the shift $\hat{Q}$; the total-variation argument of Corollary 1 gives the second claim.

*Proof (Lemma 1).* The ratio for fold $j$ is fitted on the fitting calibration families and the target families outside fold $j$, rescaled to mean one over the fitting calibration families and clipped by a fixed rule; no other family enters it. If all families are mutually independent (as under A2, and in Corollary 1 for any test law $Q$), $\hat{w}_j$ is independent of the calibration families used in (2) and (3) and of each family of fold $j$, which, conditionally on $\hat{w}_j$, are still i.i.d. $P$ and $Q$. Each query family of fold $j$ plays the role of $G_{K+1}$ in Corollary 1 and Proposition 2. If the non-training families are exchangeable (a random split of an exchangeable pool, given the training families), then conditionally on the fitting calibration families and the target families outside fold $j$, each in their given order (the classifier depends on which families are of which class and, through its cross-validation folds, on their order), $\hat{w}_j$, hence $\Phi$, is fixed, and the other calibration families together with any one family of fold $j$ are exchangeable; step (b) of the proof of Theorem 1 with $f \equiv 1$ applies directly to them, and (d) is unchanged. A unit test in the released code checks that a family's weight does not change when the other calibration families change.

*Proof (Proposition 3).* Fix $y$ and let $\mathcal{I}_y = \{k \leq K + 1 : N_{k,y} > 0\}$. Conditionally on $\mathcal{I}_y$, with $K + 1 \in \mathcal{I}_y$, the families indexed by $\mathcal{I}_y$ are exchangeable under A1 (the event $\{\mathcal{I}_y = I\}$ is invariant under permutations of $I$) and independent in the other settings, with laws $P_y = P(\cdot \mid N_y > 0)$ for calibration families and the test law conditioned on $N_y > 0$. Because $\mathbf{1}\{N_y > 0\}$ is a function of $G$, a test law with density $f(D)/\mathbb{E}_P f$ with respect to $P$ has, after conditioning, density proportional to $f(D)$ with respect to $P_y$. The label-$y$ procedures are the procedures above applied to these families with $F_k$ replaced by the label-$y$ score distribution $F_k^y$ (weights $1/N_{k,y}$), so every proof applies verbatim, and the normalizing constants cancel.

*Proof (Lemma 2).* $\mathbf{1}\{Y \in C'(X)\} \geq \mathbf{1}\{Y \in C(X)\}$ pointwise, so every average and expectation of the indicators is at least as large for $C'$.

## B Protocol Details

*Predictors.* The five unsupervised predictors are all the non-clinical models in ProteinGym v1.3's clinical score files (columns ESM1b, EVE, GEMME, TranceptEVE_L, PoET; PoET is the 200M model). The other columns are clinical predictors trained or tuned on clinical labels.

*Families and homology.* MMseqs2 release 18-8cc5c. Clusters: `cluster` with minimum identity 0.3 (0.5 for the sensitivity check), coverage 0.8 and coverage mode 0 (both sequences). Homology search: all against all, sensitivity 7.5, $E \leq 10$, at most 500 hits per query; hits are then filtered at $E \leq 10^{-3}$. Homology components additionally require identity $\geq 0.3$ and coverage $\geq 0.5$ of both sequences; the strictest definition takes connected components of all hits with $E \leq 10^{-3}$ (1,024 families, 778 single proteins, one component of 867 proteins). That component holds 45% of the benchmark's variants and, across the ten S1 draws, falls in training three times, in calibration four times and in the test set three times; family-level calibration and family-weighted coverage give it the weight of one family, whereas it dominates variant-level calibration whenever it calibrates and variant-weighted metrics whenever it is tested.

*Descriptors.* Per protein: maximum identity to any training protein over hits with $E \leq 10^{-3}$ (0 without a hit); $\log(1 + N_{\text{eff}})$ with $N_{\text{eff}}$ the sum of ProteinGym's MSA sequence weights; $\log(N_{\text{eff}}/L + 10^{-3})$ with $L$ the alignment length (written $\log N_{\text{eff}}$ per position); cosine distance of the mean-pooled last-layer (layer 12) embedding of `esm2_t12_35M_UR50D` to the nearest training protein (sequences over 1022 residues are embedded in windows of 1022 with stride 511 and residue representations averaged over windows); log length. Per family: the maximum of maximum identity and the minimum of embedding distance over member proteins, the mean of the other protein descriptors, and, in the transductive run only, log of the family's number of scored variants. Descriptors are recomputed for every split because they depend on its training families.

*Splits.* S2 uses $z$, the mean of the standardized maximum identity and $\log(1 + N_{\text{eff}})$ over non-training families, and Efraimidis–Spirakis keys $\log(u)\, e^{\beta z}$ with uniforms $u$ shared across $\beta$, so the sweep is coupled and $\beta = 0$ is S1. S3 draws test families uniformly from orphan families and training and calibration families uniformly from the rest. S4 regions are fixed once, before any S4 run: each family's embedding is the mean of its members' unit-normalized ESM2 embeddings, renormalized; after centering, the first 50 principal components are clustered by $k$-means ($k = 20$, ten restarts, seed 0). Each seed permutes the regions and adds whole regions to the test set until it holds 30% of families (the last region is split at random); training and calibration families are drawn uniformly from the remaining families. S4s permutes the regions in the same way but adds a region only if the test set stays within 30% of families, skipping the others, so that every test region is held out whole (590–624 test families per seed); training and calibration families are drawn uniformly from the remaining families in the proportions 40:30. S5 keeps the S1 training families; $z$ is the standardized mean over the five predictors of each family's within-family AUC (families without both classes get the median), and test families are drawn as in S2 with $\beta = 1$. The ten seeds resplit one fixed pool of families, so they are not independent samples of the population of families.

*Base model and conformal methods.* The base model standardizes its feature (the raw score clipped to the training 0.1% and 99.9% quantiles; in the transductive run also the score standardized over the protein's scored variants) and fits $L_2$-regularized logistic regression ($C = 1$). Earlier versions of this work (in the public repository) used the transductive base model as the primary configuration; it is now the deployment-realistic one, as in both external evaluations. Dunn et al.'s repeated subsampling uses 50 subsamples. GWCP uses nine bins: maximum identity zero, at most the median of the positive values, or above it, crossed with $\log(1 + N_{\text{eff}})$ terciles over calibration and test families; its target proportions are the bin frequencies over all test families (label-free), and it pools at the family level like HCP. For $\hat{w}$, one third of the calibration families fit the classifier (logistic regression with ten candidate penalties, five-fold cross-validated log loss) and the test families are split into five folds. ProtCOVER uses the features $(1, \hat{w}_j)$ on the remaining calibration families for queries of fold $j$; ProtCOVER-D uses an intercept and maximum identity to training, $\log(1 + N_{\text{eff}})$, $\log N_{\text{eff}}$ per position and embedding distance, standardized over all calibration families. Both feature classes

were fixed in the configuration before the conditional methods were first run. They are finite-dimensional and linear because for such classes the guarantee needs no approximation; regularized or kernel classes [18] would hold only up to a regularization term, and we do not study them. The alignment-free run removes $\log(1 + N_{\text{eff}})$ and $\log N_{\text{eff}}$ per position from the density ratio, from ProtCOVER-D's features and from the gate, and GWCP crosses maximum identity with embedding-distance terciles instead. The linear programs are solved with HiGHS 1.11 (`highspy`), warm-started across test families, with a cold re-solve and the exact simplex of Appendix A as fallbacks.

*Threshold audit.* Across the 14 runs that use conditional calibration, 17,866,020 thresholds were computed: 17,821,146 were certified from the solver's basis, 16,767 after a cold re-solve and 1,186 by the exact simplex, and 26,921 are infinite, each confirmed exactly; no unboundedness verdict of the solver was overturned, and no computation reached the pivot cap. Sets include a label when its score is at most the threshold plus $10^{-9}$, a guard against rounding of the stored scores that can only enlarge sets (Lemma 2). As an independent check of the implementation, `scripts/audit_thresholds.py` recomputes 1,800 of them independently (ProtCOVER and ProtCOVER-D; seed 0; S1, S2 at $\beta \in \{0.5, 1, 2\}$, S3 and S4; all five predictors; both classes; three random test families per fold): on the primal program with the exact test weight $\tau$, it minimizes $L$ and then maximizes the query's fitted value subject to $L(b) \leq L^\star$, both with tolerances of $10^{-10}$. Every reference computation succeeded, 25 of them only after rounding forced a relative slack of $10^{-12}$ on $L^\star$. The two computations agree on whether the threshold is infinite (both are, in 4 cases). The production threshold exceeds the reference by more than $10^{-6}$ in 7 cases, by at most $1.5 \times 10^{-1}$; there $\tau + 10^{-4}$ crosses a breakpoint of the optimal value, because with $\tau + 10^{-6}$ the same thresholds match the reference to $4.4 \times 10^{-7}$ (with $\tau + 10^{-5}$, all but one of them). It is below the reference in 23 cases, by at most $6.8 \times 10^{-8}$: 22 of them needed slack, which can raise the reference's fitted value, and the remaining one is below by $1.9 \times 10^{-9}$, within the reference's tolerances, so all are within the precision of the check. Elsewhere the two agree to $1.6 \times 10^{-9}$. No audited test variant changes membership between the two thresholds (0 lost, 0 added).

*External test set.* ClinVar `variant_summary` (October 2026) and `hgvs4variation` map VariationIDs to protein changes in the benchmark's RefSeq proteins; we keep single-nucleotide missense variants on GRCh38 with at least one review star, labeled pathogenic/likely pathogenic or benign/likely benign, whose VariationID is absent from the June 2023 release and whose protein change is not in the benchmark. Excluded as a precaution: VariationIDs present in the July 2023 release and changes that were in the June 2023 release under another VariationID. ESM-1b scores are the precomputed log-likelihood ratios of Brandes et al. [9], oriented as in the benchmark; for proteins of at most 1,022 residues they equal the benchmark's to rounding, so only those are used (longer proteins hold 43% of the new variants). The build script, the pre-registration and its amendment, and the outcome are in the repository (`docs/external_results.md`). The pre-registration was written after a sanity check of ESM-1b's AUC on these data, and the evaluation was rerun, with identical results, after each of the two fixes to the threshold computation (certification, then the exact simplex). All four hypotheses were supported in the primary and the secondary analysis: Mondrian split under-covers pathogenic variants per family in S1 (H1; 0.873 [0.864, 0.883]), HCP covers both classes at 0.89 or more (H2; 0.939 and 0.918), ProtCOVER-D's shortfall over the shifted settings is at most HCP's and GWCP's (H3; ProtCOVER was added to H3 after development had singled it out, though before the run), and abstention does not lower the error of retained singleton sets (H4, whose decision rule we fixed after the run, so it is exploratory). ESM-1b separates these new variants better than the benchmark variants of the same proteins (per-protein AUC 0.930 against 0.899), as expected if recent classifications used computational evidence (PP3/BP4), a caveat stated in the pre-registration. The higher AUC is not confined to the classifications where such evidence weighs most (exploratory, among proteins with both labels): it is 0.931 for "likely" classifications (834 proteins) and 0.934 for definitive ones (103), and 0.931 for single-submitter variants against 0.962 with several submitters or expert review (45), so these data cannot tell how much computational evidence contributes. An exploratory rerun, not pre-registered, keeps proteins of any length and scores the benchmark variants with Brandes et al.'s values as well: the external test sets grow from 4,050 to 7,117 variants on average, and no verdict changes (Mondrian split 0.870 in S1; over the shifted settings HCP falls short by 0.006 and no shift-aware method falls short).

*New-protein test set.* The same ClinVar files and variant filters, without a date filter, in approved HGNC protein-coding genes outside the benchmark (excluded by gene symbol, GeneID and UniProt accession). Each gene contributes the canonical UniProt entry that Brandes et al. scored, reached through a RefSeq protein with the same numbering, and the wild type is checked at every position. Families are MMseqs2 clusters of the new proteins with the benchmark's parameters; every new protein is also searched against every benchmark protein, which gives its maximum identity to any training set and defines the secondary population (no hit with $E \leq 10^{-3}$). Benchmark variants without a Brandes et al. score (2,755) are dropped from training and calibration, so that one score function covers both sides at any length. Clustering the new proteins jointly with the benchmark would put 1,581 of them (656 families) into a family with a benchmark protein; as pre-registered they stay in the primary population, and the secondary population excludes every family with a detectable benchmark homolog. The second predictor, ESM2 (`esm2_t30_150M_UR50D`), is scored by the wild-type marginal log-likelihood ratio [31], one forward pass per window of 1,022 residues with stride 511 and log-probabilities averaged over the windows that cover a position, negated so that higher means more pathogenic, on the same variants. Before the evaluation, as pre-registered, its per-protein mean AUC on the benchmark variants that train and calibrate it was checked: 0.808 over 2,423 proteins (ESM-1b 0.892), above the 0.5 that would have meant an orientation or position bug. The first ESM-1b run used the code before threshold certification; it was rerun after that fix and again after the exact simplex, with identical results, and the first run's outputs are in the repository with the build script and both pre-registrations.

*Gate.* The gate acts on the family of a query protein, so all variants of that family share one decision. With standardized support descriptors $\tilde{d}$, the kernel weights $\kappa_{tk} = \exp(-\|\tilde{d}_k - \tilde{d}_t\|^2/2b^2)$ give the effective number of calibration families that resemble query family $t$, $\mathrm{ESS}_t = (\sum_k \kappa_{tk})^2 / \sum_k \kappa_{tk}^2$, and the test-point mass $\pi_t = p_{K+1}$ of ProtCOVER-W measures how much of the weighted quantile's mass sits on the query itself. The gate abstains when $\mathrm{ESS}_t < m$ or $\pi_t > \pi_{\max}$: the first catches queries with few similar calibration families, the second queries that the ratio marks as far more typical of the targets than of the calibration families. Tuned under the 10% budget (a cap), it abstained on at most 1.4% of test families except at $\beta = 2$ (0.1%–5.2%).

*Gate tuning.* An inner split of each seed's training families holds out 40% as inner training families (descriptors are recomputed relative to them) and draws 30% as inner test families with the S2 sampler at $\beta = 1$; the rest are inner calibration families. Grids: kernel scale $\{0.1, 0.2, 0.35, 0.5, 1\}$ times the median pairwise distance of standardized support descriptors, $m \in \{0, 2, 5, 10, 20, 40\}$, $\pi_{\max} \in \{1, 0.05, 0.02, 0.01, 0.005\}$, abstention budget 10%, coverage tolerance 0.005.

*Metrics.* Coverage metrics average per-family coverage over families; set size and singleton rate average over variants. The under-covered fraction uses overall per-family coverage (both classes) for families with at least ten variants, for marginal and class-conditional methods alike; its binomial reference is the expected fraction under Binomial$(N_k, 1-\alpha)$. Bootstrap intervals use 1000 resamples of test families within each seed, holding that seed's calibration draw fixed, so they are conditional on the splits; seed-to-seed variation is reported separately. Random abstention in risk-coverage curves is averaged over 100 random family orders per cell. The 0.01 margin was pre-registered for the external evaluations; its placement between two noise levels of the benchmark (Section 6.2) was computed afterwards, on development data. With half or twice the margin, ProtCOVER's shortfall is smaller than GWCP's in 11 and 3 of the 25 shifted cells and larger in 0 and 0 (against HCP: 19 and 7, larger in 0 and 0). The 25 cells and three comparisons carry no multiplicity correction, and seeds resplit one pool, so these counts describe this benchmark rather than test hypotheses.

## C Synthetic Validation

We simulate families with a descriptor $d \sim N(0, 1)$, random offsets, and slopes on the score whose mean depends on $d$, with a fixed base model, so that the truth is known (300 replicates, Monte Carlo standard error of every coverage at most 0.002). Under A1 with larger families being easier, variant-level split conformal has variant-weighted coverage 0.900 but family-weighted coverage 0.831, while HCP attains 0.906 and single

subsampling 0.902. Under exact family-level covariate shift (A2, test descriptors $N(-\beta, 1)$), HCP drops to 0.826 at $\beta = 1$ and 0.734 at $\beta = 2$; weighting with the true ratio attains 0.911 and 0.950, and with cross-fitted estimated ratios 0.916 and 0.951 (class-conditional pathogenic: 0.916 and 0.947). The excess at $\beta = 2$ is the price of the test-point mass, which grows when few calibration families resemble the test family.

For the class-conditional versions we report the worse of the two class-wise coverages (target 0.90). Under the Gaussian shift at $\beta = 1$ and $\beta = 2$, it is 0.818 and 0.714 for HCP; conditional calibration on $(1, w)$ with the true ratio, the case Theorem 1 covers exactly, attains 0.909 and 0.922; ProtCOVER, with the cross-fitted ratio, 0.913 and 0.939; ProtCOVER-D, whose features $(1, d)$ do not span this ratio, 0.906 and 0.921; and reweighting with the true ratio 0.910 and 0.948. To test the theorem where the descriptor features are exact, a second shift draws calibration descriptors from $U(-2, 2)$ and test descriptors with density $(2 - d)/8$ on $[-2, 2]$, so that $w(d) = (2 - d)/2$ is affine in $d$: HCP attains 0.859, and ProtCOVER-D 0.905, conditional calibration on the true ratio 0.905, ProtCOVER 0.906 and reweighting with the true ratio 0.904. The central checks are also unit tests in the released code.

## D Additional Results

*Robustness runs.* With the transductive base model, Mondrian split still under-covers a class per family in S1 for every predictor (worse class 0.749–0.837; for TranceptEVE the benign class), and the mean shortfall under shift is 0.001 for ProtCOVER, 0.007 for GWCP and 0.023 for HCP. Without alignment depth, GWCP falls short as much as HCP (both 0.019) and ProtCOVER by 0.011. In ablations, reweighting variants instead of families leaves a mean worse-class shortfall of 0.054 over the shifted cells (pathogenic coverage in Table 4). The strictest family definition leaves no test protein with a detectable homolog, even through a shared domain, among the training or calibration proteins.

*Pooling, feature classes, information and the ratio's share.* Over the shifted cells, conditional calibration on one random variant per family and class falls short by 0.002 on average (worst cell 0.012), against 0.001 (0.008) for ProtCOVER, which pools every variant. ProtCOVER's worse-class coverage is higher with a paired interval above zero in 12 cells and lower in 2, and its pathogenic coverage varies less across seeds (s.d. 0.016 against 0.020); at matched coverage both need the same set size (1.267 against 1.268), and beyond the margin pooling is better in 1 of the 25 cells and worse in 0. Adding the descriptors to $(1, \hat{w})$ lowers the shortfall to 0.000 (worst 0.002) at a larger set size (1.316 against 1.294), and quadratic descriptor features lower ProtCOVER-D's shortfall from 0.004 to 0.001. Two comparisons separate the method from the information it uses. GWCP whose groups are the ninths of ProtCOVER's ratio (for each test fold, over the calibrating families and the fold's test families, with target proportions from the fold's test families) falls short by 0.006 (worst 0.019), against 0.007 for GWCP on its nine descriptor bins; conditional calibration on GWCP's information, a constant and the indicators of its bins (every function constant on the bins), falls short by 0.000 (worst cell 0.000) but abstains on up to 13.4% of a cell's test families (ProtCOVER: 0.3%), in the 5 cells of S2 at $\beta = 2$, because by the boundedness condition of Section 4 a bin needs at least $\tau/(1 - \tau) = 9$ calibration families with variants of a class for a finite threshold at $\alpha = 0.1$; on the families it does not abstain on, its worse-class coverage in the worst cell is 0.900, so abstention alone does not close the gap. With ProtCOVER's information, GWCP comes no closer to ProtCOVER, whose shortfall is still smaller beyond the margin in 4 cells (as many as against GWCP with its own bins) and larger in none, while with GWCP's information conditional calibration falls short less than GWCP: the gain comes mainly from the method, not from the ratio's information, and conditioning on the single feature $\hat{w}$ rather than on nine bin indicators keeps abstention rare. Fitting the ratio on 20% or 50% of the calibration families instead of a third gives ProtCOVER a shortfall of 0.001 and 0.000, within the margin of the default in every cell (Table 6).

Tables 7–10 give paired differences between ProtCOVER and HCP, GWCP and ProtCOVER-W (class-conditional versions, $\alpha = 0.1$, mean over seeds) with 95% paired family-bootstrap intervals, which are conditional on the splits. Singleton error is the fraction of single-label sets that exclude the true label.

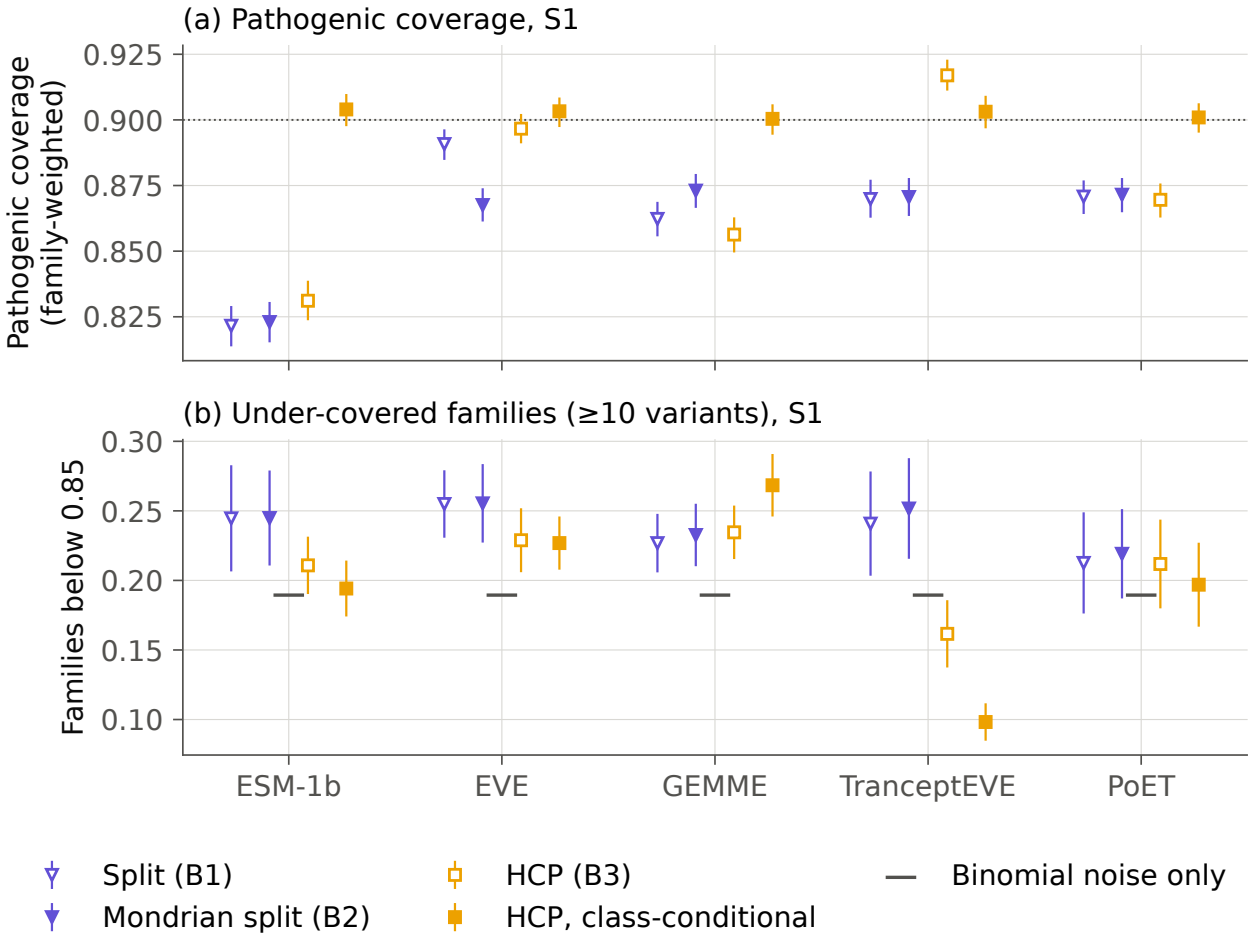



**Fig. 1.** Exchangeable families (S1, $\alpha = 0.1$). (a) Family-weighted pathogenic coverage $\mathrm{Cov}_1$ with 95% family-bootstrap intervals; the dotted line is the target. (b) Fraction of families with at least ten variants whose coverage is below 0.85 (mean and s.d. over seeds); the gray dash is the expected fraction if every family had coverage exactly 0.90.

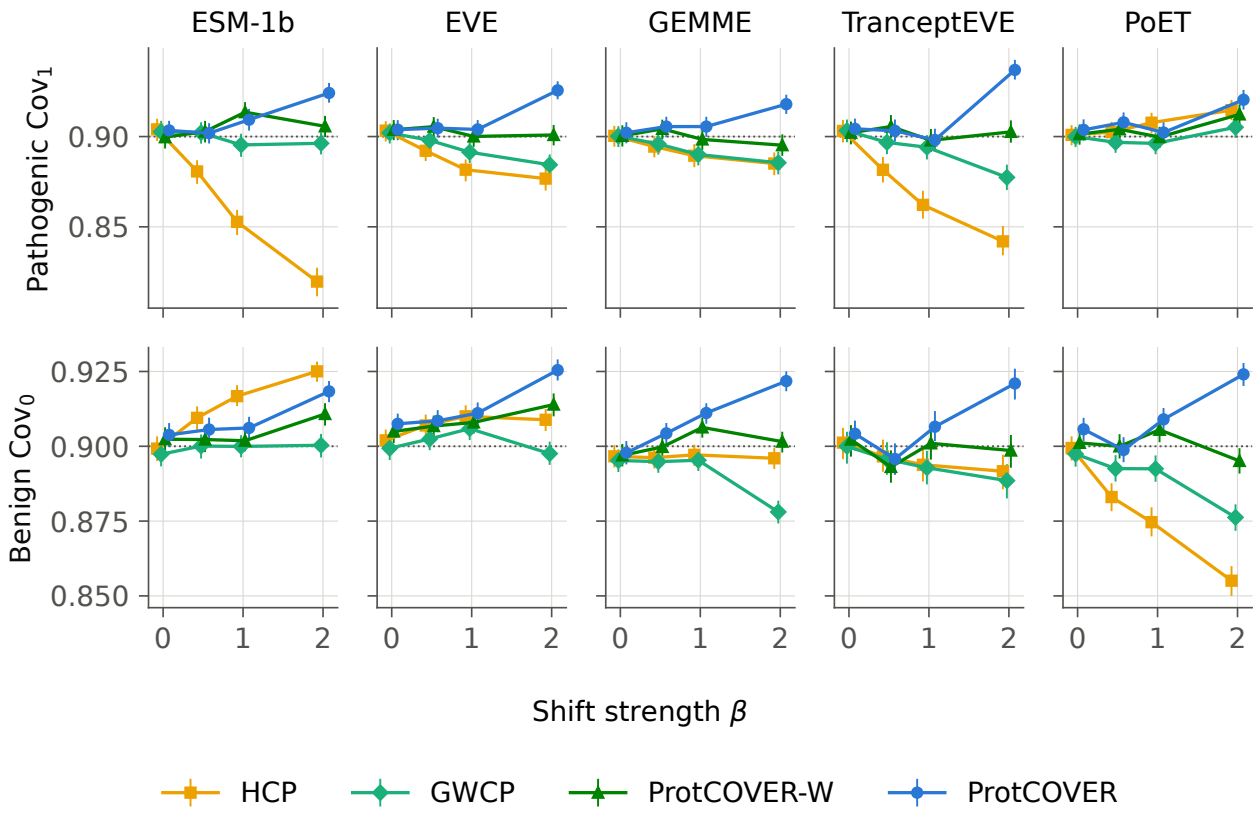



**Fig. 2.** Class-conditional coverage of both classes under increasing descriptor shift (S2; $\beta = 0$ is S1), $\alpha = 0.1$, with 95% family-bootstrap intervals; methods are offset slightly in $\beta$. Mondrian split conformal, far below target, is omitted (Fig. 1).

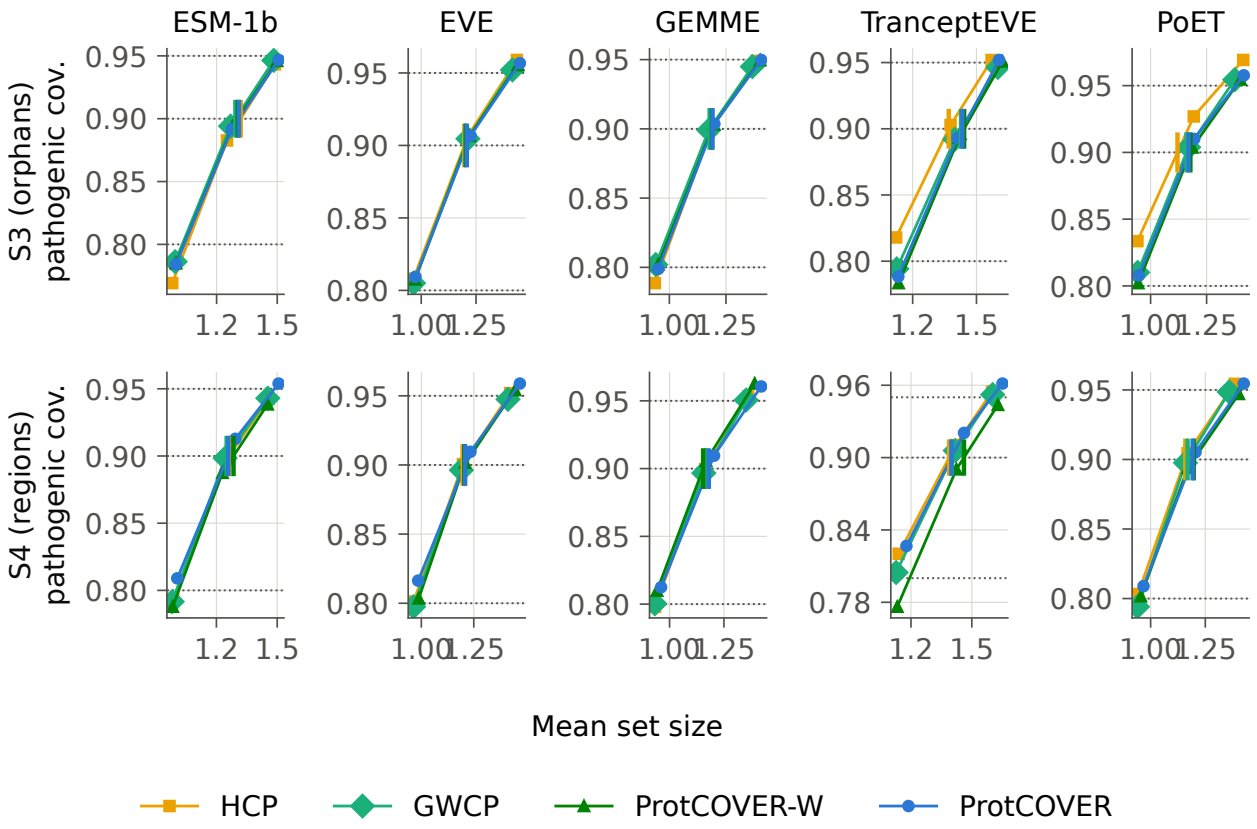



**Fig. 3.** Pathogenic coverage against mean set size for $\alpha \in \{0.2, 0.1, 0.05\}$ (left to right along each curve), class-conditional methods, on the two shifts defined without the descriptors. Dotted lines mark the three targets. A short vertical bar on the 0.90 line marks where a method's curve crosses pathogenic coverage 0.90, found by linear interpolation between its two neighboring levels: the set size the method needs for that coverage, which the last row of Table 2 averages over all shifted cells, S2 included. The methods lie close to one coverage-size curve, so a method that covers more also returns larger sets; the exceptions are HCP, above the curve for TranceptEVE and PoET in S3, and ProtCOVER-W, below it for TranceptEVE in S4.

**Table 3.** Family-weighted pathogenic / benign coverage ($\mathrm{Cov}_1$ / $\mathrm{Cov}_0$) of class-conditional methods at $\alpha = 0.1$ (target 0.90), per predictor, mean over ten seeds.

| Method | ESM-1b | EVE | GEMME | TranceptEVE | PoET |
|---|---|---|---|---|---|
| *S1 (exchangeable)* | | | | | |
| Mondrian split (B2) | 0.823 / 0.914 | 0.868 / 0.914 | 0.873 / 0.921 | 0.871 / 0.802 | 0.872 / 0.906 |
| HCP (B3) | 0.904 / 0.899 | 0.903 / 0.902 | 0.900 / 0.897 | 0.903 / 0.901 | 0.901 / 0.899 |
| GWCP (B4) | 0.903 / 0.897 | 0.902 / 0.899 | 0.900 / 0.895 | 0.903 / 0.900 | 0.900 / 0.897 |
| ProtCOVER-W | 0.900 / 0.902 | 0.904 / 0.905 | 0.900 / 0.897 | 0.902 / 0.902 | 0.901 / 0.901 |
| ProtCOVER-D | 0.910 / 0.906 | 0.907 / 0.908 | 0.906 / 0.902 | 0.910 / 0.906 | 0.909 / 0.902 |
| **ProtCOVER** | 0.903 / 0.904 | 0.904 / 0.908 | 0.902 / 0.898 | 0.904 / 0.904 | 0.904 / 0.906 |
| *S2,* $\beta = 0.5$ | | | | | |
| Mondrian split (B2) | 0.792 / 0.924 | 0.852 / 0.918 | 0.868 / 0.919 | 0.851 / 0.808 | 0.876 / 0.895 |
| HCP (B3) | 0.881 / 0.910 | 0.892 / 0.907 | 0.894 / 0.896 | 0.882 / 0.897 | 0.902 / 0.883 |
| GWCP (B4) | 0.902 / 0.900 | 0.898 / 0.902 | 0.896 / 0.895 | 0.897 / 0.896 | 0.897 / 0.893 |
| ProtCOVER-W | 0.903 / 0.902 | 0.905 / 0.907 | 0.904 / 0.900 | 0.906 / 0.893 | 0.904 / 0.900 |
| ProtCOVER-D | 0.909 / 0.906 | 0.905 / 0.908 | 0.902 / 0.904 | 0.904 / 0.904 | 0.905 / 0.900 |
| **ProtCOVER** | 0.902 / 0.906 | 0.905 / 0.909 | 0.906 / 0.904 | 0.903 / 0.896 | 0.908 / 0.899 |
| *S2,* $\beta = 1$ | | | | | |
| Mondrian split (B2) | 0.767 / 0.931 | 0.843 / 0.921 | 0.863 / 0.919 | 0.838 / 0.811 | 0.880 / 0.888 |
| HCP (B3) | 0.853 / 0.917 | 0.882 / 0.910 | 0.889 / 0.897 | 0.862 / 0.894 | 0.908 / 0.875 |
| GWCP (B4) | 0.895 / 0.900 | 0.891 / 0.906 | 0.890 / 0.895 | 0.894 / 0.893 | 0.896 / 0.892 |
| ProtCOVER-W | 0.913 / 0.902 | 0.900 / 0.908 | 0.898 / 0.906 | 0.898 / 0.901 | 0.900 / 0.906 |
| ProtCOVER-D | 0.907 / 0.910 | 0.896 / 0.912 | 0.893 / 0.906 | 0.895 / 0.907 | 0.907 / 0.901 |
| **ProtCOVER** | 0.909 / 0.906 | 0.904 / 0.911 | 0.906 / 0.911 | 0.898 / 0.907 | 0.902 / 0.909 |
| *S2,* $\beta = 2$ | | | | | |
| Mondrian split (B2) | 0.744 / 0.939 | 0.835 / 0.922 | 0.862 / 0.916 | 0.833 / 0.813 | 0.889 / 0.874 |
| HCP (B3) | 0.820 / 0.925 | 0.877 / 0.909 | 0.885 / 0.896 | 0.842 / 0.892 | 0.915 / 0.855 |
| GWCP (B4) | 0.896 / 0.900 | 0.884 / 0.898 | 0.885 / 0.878 | 0.877 / 0.889 | 0.905 / 0.876 |
| ProtCOVER-W | 0.906 / 0.911 | 0.901 / 0.914 | 0.895 / 0.902 | 0.903 / 0.899 | 0.912 / 0.895 |
| ProtCOVER-D | 0.907 / 0.918 | 0.892 / 0.908 | 0.885 / 0.906 | 0.884 / 0.908 | 0.917 / 0.886 |
| **ProtCOVER** | 0.924 / 0.918 | 0.926 / 0.925 | 0.918 / 0.922 | 0.937 / 0.921 | 0.920 / 0.924 |
| *S3 (orphan families)* | | | | | |
| Mondrian split (B2) | 0.785 / 0.926 | 0.867 / 0.911 | 0.866 / 0.933 | 0.883 / 0.777 | 0.900 / 0.899 |
| HCP (B3) | 0.883 / 0.911 | 0.907 / 0.903 | 0.900 / 0.911 | 0.903 / 0.886 | 0.927 / 0.890 |
| GWCP (B4) | 0.894 / 0.906 | 0.905 / 0.904 | 0.899 / 0.905 | 0.892 / 0.909 | 0.904 / 0.904 |
| ProtCOVER-W | 0.894 / 0.912 | 0.907 / 0.907 | 0.903 / 0.913 | 0.893 / 0.919 | 0.904 / 0.914 |
| ProtCOVER-D | 0.900 / 0.912 | 0.907 / 0.907 | 0.904 / 0.913 | 0.897 / 0.914 | 0.914 / 0.908 |
| **ProtCOVER** | 0.892 / 0.913 | 0.907 / 0.908 | 0.904 / 0.915 | 0.893 / 0.914 | 0.910 / 0.910 |
| *S4 (held-out regions)* | | | | | |
| Mondrian split (B2) | 0.810 / 0.920 | 0.862 / 0.911 | 0.877 / 0.928 | 0.896 / 0.782 | 0.876 / 0.908 |
| HCP (B3) | 0.897 / 0.902 | 0.901 / 0.901 | 0.900 / 0.904 | 0.910 / 0.887 | 0.904 / 0.902 |
| GWCP (B4) | 0.898 / 0.902 | 0.896 / 0.900 | 0.897 / 0.901 | 0.906 / 0.894 | 0.898 / 0.906 |
| ProtCOVER-W | 0.887 / 0.905 | 0.902 / 0.899 | 0.906 / 0.895 | 0.890 / 0.915 | 0.894 / 0.913 |
| ProtCOVER-D | 0.890 / 0.914 | 0.909 / 0.905 | 0.911 / 0.907 | 0.889 / 0.904 | 0.906 / 0.915 |
| **ProtCOVER** | 0.913 / 0.905 | 0.910 / 0.904 | 0.910 / 0.909 | 0.920 / 0.906 | 0.905 / 0.917 |

**Table 4.** Ablations of ProtCOVER-W, the reweighting variant: pathogenic coverage $\mathrm{Cov}_1$ at $\alpha = 0.1$, mean (minimum) over the five predictors, class-conditional versions. The clipping level $c = 20$ dates from its first version, before the conditional methods existed; ProtCOVER's own descriptor set and clipping are not ablated.

| Configuration | S2, $\beta = 1$ | S2, $\beta = 2$ | S3 | S4 |
|---|---|---|---|---|
| *Components* | | | | |
| (i) family-level calibration (HCP) | 0.879 (0.853) | 0.868 (0.820) | 0.904 (0.883) | 0.902 (0.897) |
| (i)+(ii), conditional on $\hat{w}$ (ProtCOVER) | 0.904 (0.898) | 0.925 (0.918) | 0.901 (0.892) | 0.912 (0.905) |
| (i)+(ii), conditional on descriptors (ProtCOVER-D) | 0.899 (0.893) | 0.897 (0.884) | 0.904 (0.897) | 0.901 (0.889) |
| (i)+(ii), reweighting by $\hat{w}$ (ProtCOVER-W) | 0.902 (0.898) | 0.903 (0.895) | 0.900 (0.893) | 0.896 (0.887) |
| (i)+(iii) gate | 0.879 (0.853) | 0.868 (0.820) | 0.904 (0.883) | 0.903 (0.897) |
| ProtCOVER-W + (iii) gate | 0.902 (0.898) | 0.906 (0.897) | 0.900 (0.894) | 0.896 (0.889) |
| (ii) without (i): variant-level weighting | 0.851 (0.821) | 0.852 (0.804) | 0.867 (0.819) | 0.862 (0.818) |
| *Descriptors for $\hat{w}$ (ProtCOVER-W)* | | | | |
| sequence similarity only | 0.876 (0.849) | 0.856 (0.805) | 0.899 (0.886) | 0.904 (0.900) |
| alignment depth only | 0.899 (0.890) | 0.899 (0.882) | 0.897 (0.879) | 0.902 (0.897) |
| embedding distance only | 0.882 (0.860) | 0.869 (0.822) | 0.904 (0.881) | 0.893 (0.883) |
| the three support groups | 0.902 (0.898) | 0.903 (0.895) | 0.900 (0.893) | 0.896 (0.888) |
| main set + predictor score summaries | 0.904 (0.895) | 0.902 (0.891) | 0.900 (0.887) | 0.894 (0.886) |
| *Weight clipping c (default 20; ProtCOVER-W)* | | | | |
| $c = 5$ | 0.901 (0.896) | 0.891 (0.877) | 0.900 (0.893) | 0.904 (0.901) |
| $c = 100$ | 0.902 (0.898) | 0.909 (0.902) | 0.900 (0.893) | 0.889 (0.882) |
| no clipping | 0.902 (0.898) | 0.909 (0.902) | 0.900 (0.893) | 0.852 (0.824) |

**Table 5.** Robustness runs: worse-class shortfall below the 0.90 target at $\alpha = 0.1$, mean (maximum) over the 25 shifted cells of each run, class-conditional versions; columns as in Table 2.

| | | | ProtCOVER | | |
|---|---|---|---|---|---|
| Run | HCP | GWCP | -W | -D | **default** |
| Transductive base model | 0.023 (0.102) | 0.007 (0.027) | 0.003 (0.014) | 0.003 (0.015) | 0.001 (0.010) |
| Alignment-free descriptors | 0.019 (0.080) | 0.019 (0.080) | 0.015 (0.068) | 0.016 (0.070) | 0.011 (0.060) |
| Families: 50% identity | 0.021 (0.092) | 0.009 (0.041) | 0.003 (0.011) | 0.002 (0.016) | 0.001 (0.014) |
| Families: homology components | 0.017 (0.078) | 0.008 (0.028) | 0.003 (0.009) | 0.001 (0.007) | 0.001 (0.010) |
| Families: any significant hit | 0.023 (0.109) | 0.009 (0.037) | 0.002 (0.010) | 0.001 (0.008) | 0.001 (0.011) |

**Table 6.** Pooling, feature classes, method against information, and the share of calibration families that fit $\hat{w}$, over the 25 shifted cells at $\alpha = 0.1$ (class-conditional versions). Shortfall: worse-class shortfall, mean (maximum) over cells. Size: mean set size. s.d. $\mathrm{Cov}_1$: seed-to-seed standard deviation of pathogenic coverage, mean over cells. Abst.: largest fraction of test families with both thresholds infinite in any cell. Margin: cells where ProtCOVER's shortfall is smaller / larger than the variant's by more than 0.01. Paired: cells where ProtCOVER's worse-class coverage is higher / lower than the variant's with a paired 95% interval that excludes zero. Variants were declared in the configuration before they were run.

| Variant | Shortfall | Size | s.d. $\mathrm{Cov}_1$ | Abst. | Margin | Paired |
|---|---|---|---|---|---|---|
| ProtCOVER, features $(1, \hat{w})$, ratio fitted on a third (default) | 0.001 (0.008) | 1.294 | 0.016 | 0.3% | - | - |
| one variant per family and class (Gibbs et al.) | 0.002 (0.012) | 1.292 | 0.020 | 0.3% | 1/0 | 12/2 |
| features $(1, \hat{w}, d)$ | 0.000 (0.002) | 1.316 | 0.016 | 0.3% | 0/0 | 0/21 |
| ratio fitted on 20% of calibration families | 0.001 (0.011) | 1.286 | 0.017 | 1.0% | 0/0 | 11/2 |
| ratio fitted on 50% | 0.000 (0.003) | 1.315 | 0.018 | 1.5% | 0/0 | 2/12 |
| ProtCOVER-D, features $(1, d)$ (default) | 0.004 (0.016) | 1.252 | 0.016 | 0.0% | 5/0 | - |
| features $(1, d, d^2)$ | 0.001 (0.006) | 1.275 | 0.016 | 0.4% | 0/0 | 7/10 |
| ProtCOVER-W, ratio fitted on a third (default) | 0.003 (0.013) | 1.262 | 0.023 | 0.0% | 1/0 | 16/1 |
| ratio fitted on 20% | 0.003 (0.015) | 1.262 | 0.023 | 0.0% | 1/0 | - |
| ratio fitted on 50% | 0.003 (0.016) | 1.276 | 0.024 | 0.0% | 1/0 | - |
| GWCP, nine descriptor bins (baseline) | 0.007 (0.024) | 1.241 | 0.020 | 0.0% | 4/0 | 21/1 |
| bins at the ninths of $\hat{w}$ (ProtCOVER's information) | 0.006 (0.019) | 1.246 | 0.022 | 0.0% | 4/0 | 22/0 |
| Conditional calibration on GWCP's bins (GWCP's information) | 0.000 (0.000) | 1.312 | 0.017 | 13.4% | 0/0 | 1/12 |

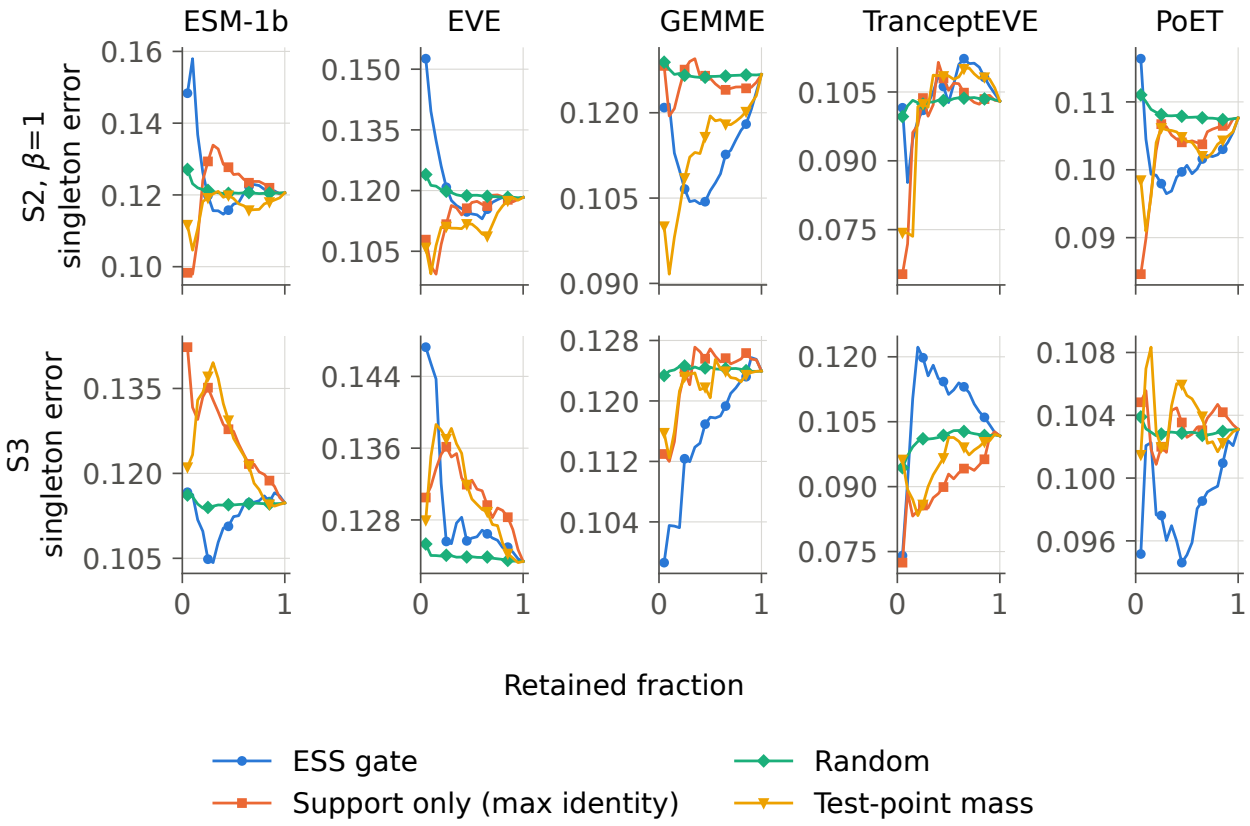



**Fig. 4.** Risk-coverage of abstention rules applied to ProtCOVER-W's class-conditional sets ($\alpha = 0.1$, mean over seeds): error of retained singleton sets as whole families are abstained in order of the score. Random abstention is averaged over 100 random family orders. The gate's decision depends only on label-free quantities, so it abstains on the same families whichever variant of (ii) it wraps.

**Table 7.** Paired differences in pathogenic coverage $\mathrm{Cov}_1$: ProtCOVER minus each baseline.

| | | $\mathrm{Cov}_1$ (pathogenic) | | |
|---|---|---|---|---|
| Setting | Predictor | − HCP | − GWCP | − ProtCOVER-W |
| S1 | ESM-1b | −0.001 [−0.004, +0.003] | +0.001 [−0.003, +0.004] | +0.004 [+0.000, +0.007] |
| | EVE | +0.001 [−0.002, +0.003] | +0.002 [−0.000, +0.004] | +0.000 [−0.002, +0.002] |
| | GEMME | +0.002 [+0.000, +0.003] | +0.002 [+0.001, +0.003] | +0.002 [+0.000, +0.003] |
| | TranceptEVE | +0.001 [−0.003, +0.006] | +0.001 [−0.003, +0.005] | +0.002 [−0.002, +0.007] |
| | PoET | +0.003 [+0.000, +0.005] | +0.004 [+0.001, +0.006] | +0.002 [+0.000, +0.005] |
| S2, $\beta = 0.5$ | ESM-1b | +0.021 [+0.017, +0.025] | −0.000 [−0.005, +0.004] | −0.001 [−0.005, +0.003] |
| | EVE | +0.013 [+0.010, +0.015] | +0.007 [+0.004, +0.009] | −0.001 [−0.003, +0.002] |
| | GEMME | +0.011 [+0.009, +0.013] | +0.010 [+0.007, +0.012] | +0.001 [−0.001, +0.003] |
| | TranceptEVE | +0.022 [+0.017, +0.026] | +0.006 [+0.002, +0.011] | −0.002 [−0.007, +0.002] |
| | PoET | +0.005 [+0.003, +0.007] | +0.011 [+0.009, +0.013] | +0.004 [+0.002, +0.006] |
| S2, $\beta = 1$ | ESM-1b | +0.057 [+0.051, +0.062] | +0.014 [+0.009, +0.019] | −0.004 [−0.009, +0.001] |
| | EVE | +0.022 [+0.019, +0.026] | +0.013 [+0.010, +0.016] | +0.004 [+0.001, +0.007] |
| | GEMME | +0.016 [+0.013, +0.019] | +0.015 [+0.013, +0.018] | +0.007 [+0.005, +0.010] |
| | TranceptEVE | +0.036 [+0.032, +0.041] | +0.004 [+0.000, +0.008] | +0.000 [−0.003, +0.004] |
| | PoET | −0.005 [−0.007, −0.004] | +0.006 [+0.004, +0.008] | +0.003 [+0.000, +0.004] |
| S2, $\beta = 2$ | ESM-1b | +0.104 [+0.098, +0.111] | +0.028 [+0.022, +0.034] | +0.018 [+0.013, +0.024] |
| | EVE | +0.049 [+0.045, +0.054] | +0.041 [+0.037, +0.046] | +0.025 [+0.021, +0.028] |
| | GEMME | +0.033 [+0.029, +0.037] | +0.032 [+0.028, +0.037] | +0.023 [+0.019, +0.027] |
| | TranceptEVE | +0.095 [+0.089, +0.102] | +0.060 [+0.054, +0.066] | +0.034 [+0.030, +0.039] |
| | PoET | +0.006 [+0.003, +0.009] | +0.015 [+0.012, +0.018] | +0.008 [+0.006, +0.011] |
| S3 | ESM-1b | +0.009 [+0.007, +0.011] | −0.002 [−0.004, −0.000] | −0.002 [−0.004, −0.000] |
| | EVE | −0.000 [−0.002, +0.001] | +0.002 [+0.000, +0.004] | +0.000 [−0.001, +0.001] |
| | GEMME | +0.004 [+0.002, +0.005] | +0.004 [+0.003, +0.006] | +0.000 [−0.001, +0.001] |
| | TranceptEVE | −0.010 [−0.012, −0.007] | +0.001 [−0.002, +0.004] | +0.000 [−0.002, +0.002] |
| | PoET | −0.017 [−0.020, −0.014] | +0.006 [+0.004, +0.009] | +0.006 [+0.004, +0.009] |
| S4 | ESM-1b | +0.016 [+0.013, +0.019] | +0.014 [+0.011, +0.018] | +0.025 [+0.022, +0.029] |
| | EVE | +0.009 [+0.007, +0.011] | +0.013 [+0.011, +0.016] | +0.008 [+0.005, +0.011] |
| | GEMME | +0.010 [+0.008, +0.012] | +0.013 [+0.010, +0.015] | +0.004 [+0.002, +0.006] |
| | TranceptEVE | +0.010 [+0.007, +0.013] | +0.015 [+0.011, +0.018] | +0.030 [+0.026, +0.035] |
| | PoET | +0.001 [−0.001, +0.003] | +0.008 [+0.005, +0.010] | +0.011 [+0.009, +0.014] |

**Table 8.** Paired differences in benign coverage $\mathrm{Cov}_0$: ProtCOVER minus each baseline.

| | | $\mathrm{Cov}_0$ (benign) | | |
|---|---|---|---|---|
| Setting | Predictor | − HCP | − GWCP | − ProtCOVER-W |
| S1 | ESM-1b | +0.005 [+0.003, +0.006] | +0.006 [+0.005, +0.008] | +0.001 [+0.000, +0.003] |
| | EVE | +0.006 [+0.004, +0.007] | +0.008 [+0.007, +0.010] | +0.003 [+0.002, +0.004] |
| | GEMME | +0.001 [−0.000, +0.002] | +0.003 [+0.001, +0.004] | +0.001 [−0.000, +0.002] |
| | TranceptEVE | +0.003 [−0.001, +0.006] | +0.004 [+0.001, +0.008] | +0.002 [−0.002, +0.005] |
| | PoET | +0.006 [+0.004, +0.008] | +0.008 [+0.006, +0.010] | +0.005 [+0.003, +0.006] |
| S2, $\beta = 0.5$ | ESM-1b | −0.004 [−0.005, −0.003] | +0.006 [+0.004, +0.007] | +0.003 [+0.002, +0.005] |
| | EVE | +0.002 [+0.001, +0.003] | +0.006 [+0.005, +0.007] | +0.002 [+0.001, +0.003] |
| | GEMME | +0.008 [+0.007, +0.009] | +0.009 [+0.008, +0.010] | +0.005 [+0.004, +0.005] |
| | TranceptEVE | −0.001 [−0.003, +0.001] | +0.000 [−0.002, +0.002] | +0.003 [+0.001, +0.005] |
| | PoET | +0.016 [+0.013, +0.018] | +0.006 [+0.004, +0.008] | −0.001 [−0.003, +0.001] |
| S2, $\beta = 1$ | ESM-1b | −0.011 [−0.012, −0.009] | +0.006 [+0.004, +0.008] | +0.004 [+0.003, +0.006] |
| | EVE | +0.001 [−0.000, +0.002] | +0.005 [+0.004, +0.006] | +0.003 [+0.002, +0.004] |
| | GEMME | +0.014 [+0.013, +0.015] | +0.016 [+0.014, +0.017] | +0.005 [+0.004, +0.006] |
| | TranceptEVE | +0.013 [+0.011, +0.015] | +0.014 [+0.012, +0.016] | +0.006 [+0.004, +0.008] |
| | PoET | +0.034 [+0.031, +0.038] | +0.017 [+0.014, +0.019] | +0.003 [+0.001, +0.006] |
| S2, $\beta = 2$ | ESM-1b | −0.007 [−0.008, −0.005] | +0.018 [+0.016, +0.020] | +0.008 [+0.006, +0.009] |
| | EVE | +0.017 [+0.015, +0.018] | +0.028 [+0.026, +0.030] | +0.012 [+0.010, +0.013] |
| | GEMME | +0.026 [+0.024, +0.028] | +0.044 [+0.041, +0.046] | +0.020 [+0.018, +0.022] |
| | TranceptEVE | +0.029 [+0.026, +0.033] | +0.032 [+0.029, +0.036] | +0.022 [+0.020, +0.026] |
| | PoET | +0.069 [+0.065, +0.073] | +0.048 [+0.044, +0.052] | +0.029 [+0.026, +0.032] |
| S3 | ESM-1b | +0.002 [+0.001, +0.003] | +0.007 [+0.006, +0.008] | +0.002 [+0.001, +0.002] |
| | EVE | +0.005 [+0.004, +0.006] | +0.003 [+0.003, +0.004] | +0.001 [+0.000, +0.001] |
| | GEMME | +0.003 [+0.002, +0.004] | +0.010 [+0.009, +0.011] | +0.001 [+0.001, +0.002] |
| | TranceptEVE | +0.028 [+0.026, +0.031] | +0.006 [+0.003, +0.008] | −0.004 [−0.006, −0.003] |
| | PoET | +0.020 [+0.018, +0.022] | +0.006 [+0.004, +0.008] | −0.004 [−0.005, −0.003] |
| S4 | ESM-1b | +0.003 [+0.002, +0.005] | +0.003 [+0.002, +0.005] | +0.001 [−0.001, +0.002] |
| | EVE | +0.003 [+0.002, +0.004] | +0.004 [+0.002, +0.005] | +0.005 [+0.004, +0.007] |
| | GEMME | +0.005 [+0.004, +0.006] | +0.008 [+0.006, +0.009] | +0.013 [+0.012, +0.015] |
| | TranceptEVE | +0.020 [+0.017, +0.023] | +0.012 [+0.009, +0.015] | −0.008 [−0.011, −0.005] |
| | PoET | +0.015 [+0.013, +0.017] | +0.011 [+0.009, +0.013] | +0.004 [+0.002, +0.006] |

**Table 9.** Paired differences in mean set size (variant-weighted): ProtCOVER minus each baseline.

| | | Mean set size | | |
|---|---|---|---|---|
| Setting | Predictor | − HCP | − GWCP | − ProtCOVER-W |
| S1 | ESM-1b | −0.002 [−0.006, +0.002] | +0.003 [−0.001, +0.007] | +0.004 [+0.000, +0.009] |
| | EVE | +0.014 [+0.012, +0.017] | +0.021 [+0.018, +0.023] | +0.007 [+0.005, +0.009] |
| | GEMME | +0.009 [+0.007, +0.011] | +0.012 [+0.009, +0.014] | +0.008 [+0.006, +0.010] |
| | TranceptEVE | −0.020 [−0.033, −0.008] | −0.018 [−0.030, −0.005] | −0.021 [−0.033, −0.008] |
| | PoET | +0.021 [+0.018, +0.025] | +0.026 [+0.022, +0.029] | +0.017 [+0.013, +0.020] |
| S2, $\beta = 0.5$ | ESM-1b | +0.033 [+0.025, +0.041] | −0.006 [−0.013, +0.001] | −0.011 [−0.018, −0.004] |
| | EVE | +0.020 [+0.017, +0.023] | +0.018 [+0.016, +0.021] | −0.002 [−0.004, +0.001] |
| | GEMME | +0.030 [+0.027, +0.033] | +0.030 [+0.028, +0.033] | +0.011 [+0.009, +0.014] |
| | TranceptEVE | +0.017 [+0.007, +0.028] | −0.010 [−0.021, +0.002] | −0.021 [−0.032, −0.009] |
| | PoET | +0.024 [+0.021, +0.027] | +0.022 [+0.020, +0.026] | −0.002 [−0.004, +0.001] |
| S2, $\beta = 1$ | ESM-1b | +0.092 [+0.082, +0.103] | +0.023 [+0.013, +0.032] | −0.018 [−0.027, −0.010] |
| | EVE | +0.037 [+0.032, +0.042] | +0.030 [+0.026, +0.035] | +0.012 [+0.008, +0.016] |
| | GEMME | +0.052 [+0.047, +0.058] | +0.053 [+0.048, +0.059] | +0.025 [+0.020, +0.030] |
| | TranceptEVE | +0.068 [+0.058, +0.080] | +0.017 [+0.005, +0.030] | +0.004 [−0.007, +0.018] |
| | PoET | +0.033 [+0.028, +0.039] | +0.032 [+0.027, +0.038] | +0.009 [+0.005, +0.014] |
| S2, $\beta = 2$ | ESM-1b | +0.207 [+0.192, +0.224] | +0.092 [+0.078, +0.106] | +0.059 [+0.049, +0.071] |
| | EVE | +0.169 [+0.156, +0.184] | +0.170 [+0.157, +0.185] | +0.114 [+0.104, +0.126] |
| | GEMME | +0.171 [+0.158, +0.187] | +0.184 [+0.170, +0.199] | +0.137 [+0.126, +0.150] |
| | TranceptEVE | +0.243 [+0.227, +0.261] | +0.191 [+0.171, +0.210] | +0.144 [+0.126, +0.164] |
| | PoET | +0.169 [+0.156, +0.184] | +0.159 [+0.145, +0.174] | +0.123 [+0.111, +0.135] |
| S3 | ESM-1b | +0.025 [+0.022, +0.028] | +0.007 [+0.004, +0.010] | −0.003 [−0.006, −0.000] |
| | EVE | +0.010 [+0.009, +0.012] | +0.012 [+0.011, +0.013] | +0.004 [+0.003, +0.005] |
| | GEMME | +0.014 [+0.012, +0.015] | +0.025 [+0.023, +0.026] | +0.003 [+0.002, +0.004] |
| | TranceptEVE | +0.033 [+0.024, +0.040] | +0.013 [+0.006, +0.019] | −0.006 [−0.011, −0.002] |
| | PoET | +0.002 [−0.000, +0.004] | +0.025 [+0.023, +0.026] | +0.008 [+0.007, +0.010] |
| S4 | ESM-1b | +0.046 [+0.036, +0.058] | +0.050 [+0.040, +0.063] | +0.064 [+0.054, +0.075] |
| | EVE | +0.036 [+0.029, +0.045] | +0.043 [+0.036, +0.052] | +0.026 [+0.020, +0.032] |
| | GEMME | +0.040 [+0.034, +0.046] | +0.047 [+0.041, +0.053] | +0.037 [+0.031, +0.042] |
| | TranceptEVE | +0.041 [+0.031, +0.051] | +0.043 [+0.034, +0.053] | +0.038 [+0.024, +0.051] |
| | PoET | +0.036 [+0.031, +0.043] | +0.046 [+0.040, +0.052] | +0.035 [+0.030, +0.040] |

**Table 10.** Paired differences in singleton error: ProtCOVER minus each baseline.

| | | Singleton error | | |
|---|---|---|---|---|
| Setting | Predictor | − HCP | − GWCP | − ProtCOVER-W |
| S1 | ESM-1b | −0.000 [−0.002, +0.001] | −0.001 [−0.003, +0.000] | +0.001 [−0.001, +0.002] |
| | EVE | −0.001 [−0.002, +0.000] | −0.003 [−0.004, −0.002] | +0.000 [−0.001, +0.002] |
| | GEMME | −0.004 [−0.004, −0.003] | −0.004 [−0.005, −0.003] | −0.003 [−0.004, −0.002] |
| | TranceptEVE | +0.011 [+0.003, +0.019] | +0.011 [+0.003, +0.019] | +0.010 [+0.002, +0.018] |
| | PoET | −0.002 [−0.004, −0.000] | −0.003 [−0.005, −0.001] | −0.001 [−0.002, +0.001] |
| S2, $\beta = 0.5$ | ESM-1b | +0.005 [+0.003, +0.008] | +0.000 [−0.002, +0.003] | +0.001 [−0.002, +0.003] |
| | EVE | −0.000 [−0.001, +0.001] | −0.001 [−0.002, +0.000] | +0.004 [+0.003, +0.005] |
| | GEMME | −0.006 [−0.006, −0.005] | −0.006 [−0.007, −0.005] | −0.002 [−0.003, −0.001] |
| | TranceptEVE | +0.015 [+0.006, +0.024] | +0.028 [+0.017, +0.038] | +0.032 [+0.020, +0.043] |
| | PoET | −0.009 [−0.010, −0.007] | −0.006 [−0.007, −0.004] | −0.001 [−0.002, +0.001] |
| S2, $\beta = 1$ | ESM-1b | +0.006 [+0.004, +0.010] | −0.001 [−0.004, +0.002] | −0.000 [−0.004, +0.003] |
| | EVE | −0.002 [−0.003, −0.001] | −0.002 [−0.003, −0.000] | +0.002 [+0.001, +0.003] |
| | GEMME | −0.010 [−0.011, −0.008] | −0.010 [−0.012, −0.009] | −0.003 [−0.005, −0.002] |
| | TranceptEVE | +0.005 [−0.003, +0.012] | +0.028 [+0.016, +0.039] | +0.028 [+0.017, +0.039] |
| | PoET | −0.016 [−0.018, −0.014] | −0.010 [−0.011, −0.008] | −0.003 [−0.005, −0.001] |
| S2, $\beta = 2$ | ESM-1b | +0.003 [−0.000, +0.007] | −0.010 [−0.013, −0.006] | −0.002 [−0.005, +0.002] |
| | EVE | −0.007 [−0.009, −0.004] | −0.011 [−0.014, −0.009] | +0.001 [−0.001, +0.003] |
| | GEMME | −0.010 [−0.013, −0.007] | −0.021 [−0.024, −0.018] | −0.005 [−0.008, −0.003] |
| | TranceptEVE | −0.005 [−0.015, +0.006] | +0.020 [+0.004, +0.035] | +0.032 [+0.018, +0.046] |
| | PoET | −0.027 [−0.030, −0.024] | −0.019 [−0.022, −0.016] | −0.011 [−0.013, −0.008] |
| S3 | ESM-1b | −0.003 [−0.003, −0.002] | −0.005 [−0.006, −0.004] | −0.002 [−0.002, −0.001] |
| | EVE | −0.002 [−0.003, −0.001] | −0.002 [−0.003, −0.002] | −0.000 [−0.001, +0.000] |
| | GEMME | −0.003 [−0.003, −0.002] | −0.006 [−0.007, −0.005] | −0.001 [−0.001, −0.000] |
| | TranceptEVE | +0.005 [−0.002, +0.013] | −0.000 [−0.007, +0.006] | +0.002 [−0.002, +0.005] |
| | PoET | −0.002 [−0.004, −0.001] | −0.003 [−0.004, −0.001] | +0.002 [+0.001, +0.003] |
| S4 | ESM-1b | +0.001 [−0.002, +0.003] | −0.001 [−0.003, +0.001] | −0.003 [−0.005, −0.001] |
| | EVE | −0.001 [−0.002, +0.001] | −0.004 [−0.006, −0.003] | −0.005 [−0.007, −0.003] |
| | GEMME | −0.004 [−0.005, −0.003] | −0.006 [−0.008, −0.005] | −0.006 [−0.007, −0.004] |
| | TranceptEVE | +0.005 [−0.004, +0.014] | −0.007 [−0.015, +0.001] | −0.032 [−0.045, −0.019] |
| | PoET | −0.004 [−0.006, −0.002] | −0.005 [−0.006, −0.003] | −0.004 [−0.006, −0.003] |

Tables 11–13 report every method, marginal (cc = no) and class-conditional (cc = yes), averaged over the five predictors: family-weighted coverage Cov, variant-weighted coverage, $\mathrm{Cov}_1$, $\mathrm{Cov}_0$, mean set size, singleton rate, the fraction of families with at least ten variants whose coverage is below $1 - \alpha - 0.05$, and the fraction of test families on which the method abstained. Per-predictor values with bootstrap intervals, and the summaries of every sensitivity, stress and external run, are in `results/` of the code repository.

Table 11: All methods at $\alpha = 0.1$, mean over predictors and seeds.

| Method | cc | Cov | var.-w. | $\mathrm{Cov}_1$ | $\mathrm{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| *S1 (exchangeable)* | | | | | | | | | |
| Split (B1/B2) | no | 0.892 | 0.896 | 0.863 | 0.893 | 1.174 | 0.826 | 0.236 | 0.000 |
| Split (B1/B2) | yes | 0.890 | 0.896 | 0.861 | 0.891 | 1.172 | 0.828 | 0.241 | 0.000 |
| HCP (B3) | no | 0.900 | 0.908 | 0.874 | 0.902 | 1.207 | 0.793 | 0.210 | 0.000 |
| HCP (B3) | yes | 0.909 | 0.914 | 0.902 | 0.900 | 1.241 | 0.759 | 0.197 | 0.000 |
| Subsample once (B3) | no | 0.899 | 0.907 | 0.874 | 0.901 | 1.204 | 0.796 | 0.215 | 0.000 |
| Subsample once (B3) | yes | 0.908 | 0.914 | 0.903 | 0.898 | 1.242 | 0.758 | 0.199 | 0.000 |
| Repeated subs. (B3) | no | 0.900 | 0.908 | 0.874 | 0.902 | 1.207 | 0.793 | 0.210 | 0.000 |
| Repeated subs. (B3) | yes | 0.908 | 0.913 | 0.902 | 0.899 | 1.240 | 0.760 | 0.199 | 0.000 |
| GWCP (B4) | no | 0.899 | 0.906 | 0.872 | 0.901 | 1.201 | 0.799 | 0.216 | 0.000 |
| GWCP (B4) | yes | 0.907 | 0.913 | 0.902 | 0.898 | 1.237 | 0.763 | 0.202 | 0.000 |
| ProtCOVER | no | 0.903 | 0.906 | 0.872 | 0.907 | 1.206 | 0.794 | 0.204 | 0.000 |
| ProtCOVER | yes | 0.910 | 0.914 | 0.903 | 0.904 | 1.246 | 0.754 | 0.196 | 0.000 |
| ProtCOVER-D | no | 0.906 | 0.897 | 0.868 | 0.914 | 1.196 | 0.804 | 0.202 | 0.000 |
| ProtCOVER-D | yes | 0.909 | 0.912 | 0.908 | 0.905 | 1.242 | 0.758 | 0.207 | 0.000 |
| ProtCOVER-W | no | 0.901 | 0.908 | 0.874 | 0.902 | 1.208 | 0.792 | 0.209 | 0.000 |
| ProtCOVER-W | yes | 0.909 | 0.914 | 0.901 | 0.902 | 1.243 | 0.757 | 0.194 | 0.000 |
| (i)+(iii) gate | no | 0.900 | 0.908 | 0.874 | 0.902 | 1.207 | 0.793 | 0.209 | 0.000 |
| (i)+(iii) gate | yes | 0.909 | 0.914 | 0.902 | 0.900 | 1.242 | 0.758 | 0.197 | 0.001 |
| ProtCOVER-W + gate | no | 0.901 | 0.908 | 0.874 | 0.903 | 1.208 | 0.792 | 0.209 | 0.001 |
| ProtCOVER-W + gate | yes | 0.910 | 0.914 | 0.901 | 0.902 | 1.243 | 0.757 | 0.194 | 0.000 |
| Support-only (B5) | no | 0.901 | 0.908 | 0.874 | 0.902 | 1.208 | 0.792 | 0.209 | 0.001 |
| Support-only (B5) | yes | 0.909 | 0.914 | 0.901 | 0.902 | 1.243 | 0.757 | 0.194 | 0.000 |
| Random abst. (B5) | no | 0.901 | 0.908 | 0.874 | 0.902 | 1.208 | 0.792 | 0.209 | 0.001 |
| Random abst. (B5) | yes | 0.909 | 0.914 | 0.901 | 0.902 | 1.243 | 0.757 | 0.194 | 0.000 |
| *S2,* $\beta = 0.5$ | | | | | | | | | |
| Split (B1/B2) | no | 0.889 | 0.892 | 0.854 | 0.892 | 1.170 | 0.830 | 0.248 | 0.000 |
| Split (B1/B2) | yes | 0.888 | 0.891 | 0.848 | 0.893 | 1.168 | 0.832 | 0.252 | 0.000 |
| HCP (B3) | no | 0.897 | 0.904 | 0.866 | 0.900 | 1.202 | 0.798 | 0.223 | 0.000 |
| HCP (B3) | yes | 0.904 | 0.908 | 0.890 | 0.898 | 1.229 | 0.771 | 0.215 | 0.000 |
| Subsample once (B3) | no | 0.896 | 0.902 | 0.866 | 0.897 | 1.198 | 0.802 | 0.232 | 0.000 |
| Subsample once (B3) | yes | 0.904 | 0.908 | 0.892 | 0.897 | 1.231 | 0.769 | 0.216 | 0.000 |
| Repeated subs. (B3) | no | 0.897 | 0.904 | 0.866 | 0.900 | 1.201 | 0.799 | 0.224 | 0.000 |
| Repeated subs. (B3) | yes | 0.904 | 0.908 | 0.890 | 0.898 | 1.229 | 0.771 | 0.215 | 0.000 |
| GWCP (B4) | no | 0.898 | 0.904 | 0.866 | 0.901 | 1.203 | 0.797 | 0.222 | 0.000 |
| GWCP (B4) | yes | 0.906 | 0.911 | 0.898 | 0.897 | 1.242 | 0.758 | 0.211 | 0.000 |
| ProtCOVER | no | 0.902 | 0.908 | 0.872 | 0.905 | 1.217 | 0.783 | 0.208 | 0.000 |
| ProtCOVER | yes | 0.910 | 0.911 | 0.905 | 0.903 | 1.253 | 0.747 | 0.198 | 0.000 |
| ProtCOVER-D | no | 0.904 | 0.896 | 0.861 | 0.913 | 1.198 | 0.802 | 0.209 | 0.000 |
| ProtCOVER-D | yes | 0.908 | 0.911 | 0.905 | 0.905 | 1.250 | 0.750 | 0.213 | 0.000 |
| ProtCOVER-W | no | 0.901 | 0.907 | 0.870 | 0.904 | 1.214 | 0.786 | 0.211 | 0.000 |
| ProtCOVER-W | yes | 0.910 | 0.915 | 0.904 | 0.900 | 1.258 | 0.742 | 0.196 | 0.000 |
| (i)+(iii) gate | no | 0.897 | 0.904 | 0.866 | 0.900 | 1.202 | 0.798 | 0.223 | 0.000 |

*All methods at $\alpha = 0.1$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| (i)+(iii) gate | yes | 0.904 | 0.908 | 0.890 | 0.899 | 1.229 | 0.771 | 0.215 | 0.001 |
| ProtCOVER-W + gate | no | 0.901 | 0.907 | 0.870 | 0.904 | 1.215 | 0.785 | 0.211 | 0.001 |
| ProtCOVER-W + gate | yes | 0.910 | 0.915 | 0.904 | 0.900 | 1.259 | 0.741 | 0.196 | 0.000 |
| Support-only (B5) | no | 0.901 | 0.907 | 0.870 | 0.904 | 1.214 | 0.786 | 0.211 | 0.001 |
| Support-only (B5) | yes | 0.910 | 0.915 | 0.904 | 0.900 | 1.259 | 0.741 | 0.196 | 0.000 |
| Random abst. (B5) | no | 0.901 | 0.907 | 0.870 | 0.904 | 1.215 | 0.785 | 0.211 | 0.001 |
| Random abst. (B5) | yes | 0.910 | 0.915 | 0.904 | 0.900 | 1.259 | 0.741 | 0.196 | 0.000 |
| *S2, $\beta = 1$* | | | | | | | | | |
| Split (B1/B2) | no | 0.887 | 0.890 | 0.847 | 0.891 | 1.168 | 0.832 | 0.257 | 0.000 |
| Split (B1/B2) | yes | 0.886 | 0.889 | 0.838 | 0.894 | 1.166 | 0.834 | 0.261 | 0.000 |
| HCP (B3) | no | 0.895 | 0.902 | 0.860 | 0.900 | 1.199 | 0.801 | 0.231 | 0.000 |
| HCP (B3) | yes | 0.901 | 0.904 | 0.879 | 0.898 | 1.219 | 0.781 | 0.226 | 0.000 |
| Subsample once (B3) | no | 0.895 | 0.902 | 0.859 | 0.900 | 1.200 | 0.800 | 0.232 | 0.000 |
| Subsample once (B3) | yes | 0.902 | 0.904 | 0.879 | 0.899 | 1.223 | 0.777 | 0.223 | 0.000 |
| Repeated subs. (B3) | no | 0.896 | 0.902 | 0.859 | 0.900 | 1.199 | 0.801 | 0.231 | 0.000 |
| Repeated subs. (B3) | yes | 0.901 | 0.904 | 0.879 | 0.898 | 1.219 | 0.781 | 0.227 | 0.000 |
| GWCP (B4) | no | 0.897 | 0.903 | 0.861 | 0.901 | 1.204 | 0.796 | 0.227 | 0.000 |
| GWCP (B4) | yes | 0.905 | 0.909 | 0.893 | 0.897 | 1.245 | 0.755 | 0.216 | 0.000 |
| ProtCOVER | no | 0.908 | 0.913 | 0.880 | 0.910 | 1.248 | 0.752 | 0.198 | 0.000 |
| ProtCOVER | yes | 0.914 | 0.914 | 0.904 | 0.909 | 1.276 | 0.724 | 0.193 | 0.000 |
| ProtCOVER-D | no | 0.903 | 0.896 | 0.855 | 0.914 | 1.201 | 0.799 | 0.214 | 0.000 |
| ProtCOVER-D | yes | 0.909 | 0.911 | 0.899 | 0.907 | 1.255 | 0.745 | 0.211 | 0.000 |
| ProtCOVER-W | no | 0.903 | 0.910 | 0.870 | 0.906 | 1.225 | 0.775 | 0.206 | 0.000 |
| ProtCOVER-W | yes | 0.912 | 0.915 | 0.902 | 0.905 | 1.269 | 0.731 | 0.195 | 0.000 |
| (i)+(iii) gate | no | 0.896 | 0.902 | 0.860 | 0.900 | 1.199 | 0.801 | 0.231 | 0.000 |
| (i)+(iii) gate | yes | 0.901 | 0.904 | 0.879 | 0.899 | 1.220 | 0.780 | 0.226 | 0.001 |
| ProtCOVER-W + gate | no | 0.903 | 0.910 | 0.871 | 0.907 | 1.225 | 0.775 | 0.206 | 0.001 |
| ProtCOVER-W + gate | yes | 0.912 | 0.915 | 0.902 | 0.905 | 1.269 | 0.731 | 0.195 | 0.000 |
| Support-only (B5) | no | 0.903 | 0.910 | 0.871 | 0.907 | 1.225 | 0.775 | 0.206 | 0.001 |
| Support-only (B5) | yes | 0.912 | 0.915 | 0.902 | 0.905 | 1.269 | 0.731 | 0.195 | 0.000 |
| Random abst. (B5) | no | 0.903 | 0.910 | 0.871 | 0.907 | 1.225 | 0.775 | 0.206 | 0.001 |
| Random abst. (B5) | yes | 0.912 | 0.915 | 0.902 | 0.905 | 1.269 | 0.731 | 0.195 | 0.000 |
| *S2, $\beta = 2$* | | | | | | | | | |
| Split (B1/B2) | no | 0.886 | 0.889 | 0.842 | 0.890 | 1.167 | 0.833 | 0.263 | 0.000 |
| Split (B1/B2) | yes | 0.884 | 0.888 | 0.833 | 0.893 | 1.165 | 0.835 | 0.269 | 0.000 |
| HCP (B3) | no | 0.894 | 0.900 | 0.854 | 0.898 | 1.196 | 0.804 | 0.239 | 0.000 |
| HCP (B3) | yes | 0.896 | 0.898 | 0.868 | 0.895 | 1.208 | 0.792 | 0.243 | 0.000 |
| Subsample once (B3) | no | 0.890 | 0.897 | 0.854 | 0.893 | 1.190 | 0.810 | 0.251 | 0.000 |
| Subsample once (B3) | yes | 0.894 | 0.896 | 0.866 | 0.892 | 1.203 | 0.797 | 0.251 | 0.000 |
| Repeated subs. (B3) | no | 0.893 | 0.900 | 0.854 | 0.898 | 1.194 | 0.806 | 0.241 | 0.000 |
| Repeated subs. (B3) | yes | 0.896 | 0.898 | 0.868 | 0.895 | 1.208 | 0.792 | 0.243 | 0.000 |
| GWCP (B4) | no | 0.894 | 0.900 | 0.855 | 0.899 | 1.201 | 0.799 | 0.242 | 0.000 |
| GWCP (B4) | yes | 0.898 | 0.902 | 0.890 | 0.888 | 1.241 | 0.759 | 0.243 | 0.000 |
| ProtCOVER | no | 0.926 | 0.931 | 0.908 | 0.927 | 1.387 | 0.613 | 0.156 | 0.000 |
| ProtCOVER | yes | 0.928 | 0.929 | 0.925 | 0.922 | 1.400 | 0.600 | 0.160 | 0.000 |
| ProtCOVER-D | no | 0.900 | 0.896 | 0.850 | 0.912 | 1.203 | 0.797 | 0.221 | 0.000 |
| ProtCOVER-D | yes | 0.907 | 0.909 | 0.897 | 0.905 | 1.270 | 0.730 | 0.220 | 0.000 |
| ProtCOVER-W | no | 0.909 | 0.915 | 0.880 | 0.912 | 1.259 | 0.741 | 0.192 | 0.000 |
| ProtCOVER-W | yes | 0.912 | 0.916 | 0.903 | 0.904 | 1.284 | 0.716 | 0.195 | 0.000 |

*All methods at $\alpha = 0.1$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| (i)+(iii) gate | no | 0.894 | 0.900 | 0.854 | 0.898 | 1.196 | 0.804 | 0.239 | 0.000 |
| (i)+(iii) gate | yes | 0.896 | 0.899 | 0.868 | 0.895 | 1.209 | 0.791 | 0.242 | 0.001 |
| ProtCOVER-W + gate | no | 0.910 | 0.916 | 0.881 | 0.912 | 1.264 | 0.736 | 0.190 | 0.007 |
| ProtCOVER-W + gate | yes | 0.914 | 0.918 | 0.906 | 0.906 | 1.301 | 0.699 | 0.191 | 0.028 |
| Support-only (B5) | no | 0.909 | 0.916 | 0.881 | 0.912 | 1.263 | 0.737 | 0.190 | 0.007 |
| Support-only (B5) | yes | 0.914 | 0.917 | 0.905 | 0.907 | 1.300 | 0.700 | 0.192 | 0.028 |
| Random abst. (B5) | no | 0.910 | 0.916 | 0.881 | 0.912 | 1.264 | 0.736 | 0.190 | 0.007 |
| Random abst. (B5) | yes | 0.914 | 0.918 | 0.905 | 0.907 | 1.302 | 0.698 | 0.191 | 0.028 |
| *S3 (orphan families)* | | | | | | | | | |
| Split (B1/B2) | no | 0.890 | 0.893 | 0.863 | 0.890 | 1.174 | 0.826 | 0.248 | 0.000 |
| Split (B1/B2) | yes | 0.888 | 0.893 | 0.860 | 0.889 | 1.171 | 0.829 | 0.251 | 0.000 |
| HCP (B3) | no | 0.899 | 0.904 | 0.875 | 0.900 | 1.209 | 0.791 | 0.219 | 0.000 |
| HCP (B3) | yes | 0.910 | 0.913 | 0.904 | 0.900 | 1.244 | 0.756 | 0.198 | 0.000 |
| Subsample once (B3) | no | 0.898 | 0.903 | 0.873 | 0.900 | 1.207 | 0.793 | 0.225 | 0.000 |
| Subsample once (B3) | yes | 0.909 | 0.912 | 0.903 | 0.899 | 1.243 | 0.757 | 0.202 | 0.000 |
| Repeated subs. (B3) | no | 0.899 | 0.904 | 0.875 | 0.900 | 1.208 | 0.792 | 0.221 | 0.000 |
| Repeated subs. (B3) | yes | 0.910 | 0.913 | 0.904 | 0.900 | 1.244 | 0.756 | 0.199 | 0.000 |
| GWCP (B4) | no | 0.902 | 0.908 | 0.880 | 0.904 | 1.226 | 0.774 | 0.210 | 0.000 |
| GWCP (B4) | yes | 0.911 | 0.911 | 0.899 | 0.905 | 1.245 | 0.755 | 0.202 | 0.000 |
| ProtCOVER | no | 0.907 | 0.912 | 0.886 | 0.908 | 1.244 | 0.756 | 0.196 | 0.000 |
| ProtCOVER | yes | 0.915 | 0.916 | 0.901 | 0.912 | 1.261 | 0.739 | 0.185 | 0.000 |
| ProtCOVER-D | no | 0.909 | 0.904 | 0.874 | 0.917 | 1.219 | 0.781 | 0.201 | 0.000 |
| ProtCOVER-D | yes | 0.912 | 0.910 | 0.904 | 0.911 | 1.244 | 0.756 | 0.208 | 0.000 |
| ProtCOVER-W | no | 0.907 | 0.912 | 0.886 | 0.909 | 1.244 | 0.756 | 0.197 | 0.000 |
| ProtCOVER-W | yes | 0.916 | 0.916 | 0.900 | 0.913 | 1.260 | 0.740 | 0.185 | 0.000 |
| (i)+(iii) gate | no | 0.899 | 0.904 | 0.875 | 0.900 | 1.209 | 0.791 | 0.219 | 0.000 |
| (i)+(iii) gate | yes | 0.910 | 0.913 | 0.904 | 0.901 | 1.245 | 0.755 | 0.198 | 0.002 |
| ProtCOVER-W + gate | no | 0.907 | 0.912 | 0.886 | 0.909 | 1.244 | 0.756 | 0.197 | 0.000 |
| ProtCOVER-W + gate | yes | 0.916 | 0.916 | 0.900 | 0.913 | 1.261 | 0.739 | 0.185 | 0.002 |
| Support-only (B5) | no | 0.907 | 0.912 | 0.886 | 0.909 | 1.244 | 0.756 | 0.197 | 0.000 |
| Support-only (B5) | yes | 0.916 | 0.916 | 0.900 | 0.913 | 1.261 | 0.739 | 0.185 | 0.002 |
| Random abst. (B5) | no | 0.907 | 0.912 | 0.886 | 0.909 | 1.244 | 0.756 | 0.197 | 0.000 |
| Random abst. (B5) | yes | 0.916 | 0.916 | 0.900 | 0.913 | 1.261 | 0.739 | 0.185 | 0.002 |
| *S4 (held-out regions)* | | | | | | | | | |
| Split (B1/B2) | no | 0.891 | 0.895 | 0.864 | 0.895 | 1.182 | 0.818 | 0.236 | 0.000 |
| Split (B1/B2) | yes | 0.889 | 0.896 | 0.864 | 0.890 | 1.177 | 0.823 | 0.242 | 0.000 |
| HCP (B3) | no | 0.898 | 0.904 | 0.874 | 0.902 | 1.208 | 0.792 | 0.216 | 0.000 |
| HCP (B3) | yes | 0.908 | 0.914 | 0.902 | 0.899 | 1.240 | 0.760 | 0.198 | 0.000 |
| Subsample once (B3) | no | 0.898 | 0.903 | 0.875 | 0.901 | 1.206 | 0.794 | 0.218 | 0.000 |
| Subsample once (B3) | yes | 0.907 | 0.913 | 0.901 | 0.899 | 1.239 | 0.761 | 0.199 | 0.000 |
| Repeated subs. (B3) | no | 0.898 | 0.903 | 0.873 | 0.902 | 1.208 | 0.792 | 0.216 | 0.000 |
| Repeated subs. (B3) | yes | 0.908 | 0.914 | 0.903 | 0.899 | 1.240 | 0.760 | 0.197 | 0.000 |
| GWCP (B4) | no | 0.898 | 0.904 | 0.875 | 0.903 | 1.210 | 0.790 | 0.216 | 0.000 |
| GWCP (B4) | yes | 0.907 | 0.910 | 0.899 | 0.901 | 1.234 | 0.766 | 0.204 | 0.000 |
| ProtCOVER | no | 0.907 | 0.911 | 0.883 | 0.912 | 1.250 | 0.750 | 0.197 | 0.003 |
| ProtCOVER | yes | 0.915 | 0.919 | 0.912 | 0.908 | 1.280 | 0.720 | 0.184 | 0.003 |
| ProtCOVER-D | no | 0.907 | 0.898 | 0.872 | 0.918 | 1.213 | 0.787 | 0.201 | 0.000 |
| ProtCOVER-D | yes | 0.908 | 0.909 | 0.901 | 0.909 | 1.241 | 0.759 | 0.212 | 0.000 |
| ProtCOVER-W | no | 0.905 | 0.910 | 0.881 | 0.910 | 1.240 | 0.760 | 0.199 | 0.000 |

*All methods at $\alpha = 0.1$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| ProtCOVER-W | yes | 0.908 | 0.906 | 0.896 | 0.905 | 1.240 | 0.760 | 0.210 | 0.000 |
| (i)+(iii) gate | no | 0.898 | 0.904 | 0.874 | 0.902 | 1.208 | 0.792 | 0.215 | 0.001 |
| (i)+(iii) gate | yes | 0.908 | 0.914 | 0.903 | 0.899 | 1.242 | 0.758 | 0.197 | 0.001 |
| ProtCOVER-W + gate | no | 0.905 | 0.910 | 0.882 | 0.910 | 1.240 | 0.760 | 0.199 | 0.001 |
| ProtCOVER-W + gate | yes | 0.908 | 0.907 | 0.896 | 0.905 | 1.243 | 0.757 | 0.209 | 0.005 |
| Support-only (B5) | no | 0.905 | 0.910 | 0.882 | 0.910 | 1.240 | 0.760 | 0.198 | 0.001 |
| Support-only (B5) | yes | 0.908 | 0.907 | 0.896 | 0.906 | 1.243 | 0.757 | 0.209 | 0.005 |
| Random abst. (B5) | no | 0.905 | 0.910 | 0.882 | 0.910 | 1.240 | 0.760 | 0.199 | 0.001 |
| Random abst. (B5) | yes | 0.908 | 0.907 | 0.896 | 0.906 | 1.243 | 0.757 | 0.209 | 0.005 |

Table 12: All methods at $\alpha = 0.05$, mean over predictors and seeds.

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| *S1 (exchangeable)* | | | | | | | | | |
| Split (B1/B2) | no | 0.941 | 0.949 | 0.925 | 0.939 | 1.372 | 0.628 | 0.149 | 0.000 |
| Split (B1/B2) | yes | 0.943 | 0.948 | 0.920 | 0.945 | 1.371 | 0.629 | 0.145 | 0.000 |
| HCP (B3) | no | 0.951 | 0.958 | 0.938 | 0.951 | 1.445 | 0.555 | 0.117 | 0.000 |
| HCP (B3) | yes | 0.956 | 0.959 | 0.952 | 0.951 | 1.454 | 0.546 | 0.115 | 0.000 |
| Subsample once (B3) | no | 0.950 | 0.956 | 0.937 | 0.950 | 1.438 | 0.562 | 0.124 | 0.000 |
| Subsample once (B3) | yes | 0.956 | 0.959 | 0.953 | 0.951 | 1.455 | 0.545 | 0.116 | 0.000 |
| Repeated subs. (B3) | no | 0.951 | 0.958 | 0.938 | 0.951 | 1.446 | 0.554 | 0.117 | 0.000 |
| Repeated subs. (B3) | yes | 0.956 | 0.959 | 0.952 | 0.951 | 1.453 | 0.547 | 0.116 | 0.000 |
| GWCP (B4) | no | 0.949 | 0.956 | 0.936 | 0.950 | 1.434 | 0.566 | 0.124 | 0.000 |
| GWCP (B4) | yes | 0.955 | 0.958 | 0.951 | 0.949 | 1.446 | 0.554 | 0.121 | 0.000 |
| ProtCOVER | no | 0.954 | 0.958 | 0.938 | 0.956 | 1.440 | 0.560 | 0.112 | 0.000 |
| ProtCOVER | yes | 0.959 | 0.961 | 0.955 | 0.954 | 1.467 | 0.533 | 0.109 | 0.000 |
| ProtCOVER-D | no | 0.956 | 0.953 | 0.933 | 0.961 | 1.424 | 0.576 | 0.110 | 0.000 |
| ProtCOVER-D | yes | 0.958 | 0.961 | 0.957 | 0.956 | 1.478 | 0.522 | 0.112 | 0.000 |
| ProtCOVER-W | no | 0.952 | 0.958 | 0.939 | 0.952 | 1.449 | 0.551 | 0.115 | 0.000 |
| ProtCOVER-W | yes | 0.957 | 0.960 | 0.953 | 0.952 | 1.461 | 0.539 | 0.111 | 0.000 |
| (i)+(iii) gate | no | 0.951 | 0.958 | 0.938 | 0.951 | 1.445 | 0.555 | 0.117 | 0.000 |
| (i)+(iii) gate | yes | 0.956 | 0.959 | 0.952 | 0.951 | 1.454 | 0.546 | 0.115 | 0.000 |
| ProtCOVER-W + gate | no | 0.952 | 0.958 | 0.939 | 0.952 | 1.449 | 0.551 | 0.115 | 0.000 |
| ProtCOVER-W + gate | yes | 0.957 | 0.960 | 0.953 | 0.952 | 1.461 | 0.539 | 0.111 | 0.000 |
| Support-only (B5) | no | 0.952 | 0.958 | 0.939 | 0.952 | 1.449 | 0.551 | 0.115 | 0.000 |
| Support-only (B5) | yes | 0.957 | 0.960 | 0.953 | 0.952 | 1.461 | 0.539 | 0.111 | 0.000 |
| Random abst. (B5) | no | 0.952 | 0.958 | 0.939 | 0.952 | 1.449 | 0.551 | 0.115 | 0.000 |
| Random abst. (B5) | yes | 0.957 | 0.960 | 0.953 | 0.952 | 1.461 | 0.539 | 0.111 | 0.000 |
| *S2, $\beta = 0.5$* | | | | | | | | | |
| Split (B1/B2) | no | 0.938 | 0.946 | 0.918 | 0.937 | 1.366 | 0.634 | 0.162 | 0.000 |
| Split (B1/B2) | yes | 0.940 | 0.945 | 0.910 | 0.945 | 1.364 | 0.636 | 0.158 | 0.000 |
| HCP (B3) | no | 0.949 | 0.956 | 0.933 | 0.949 | 1.441 | 0.559 | 0.127 | 0.000 |
| HCP (B3) | yes | 0.953 | 0.957 | 0.947 | 0.948 | 1.447 | 0.553 | 0.130 | 0.000 |
| Subsample once (B3) | no | 0.947 | 0.955 | 0.931 | 0.948 | 1.437 | 0.563 | 0.131 | 0.000 |
| Subsample once (B3) | yes | 0.952 | 0.955 | 0.946 | 0.947 | 1.441 | 0.559 | 0.133 | 0.000 |
| Repeated subs. (B3) | no | 0.949 | 0.956 | 0.934 | 0.950 | 1.441 | 0.559 | 0.127 | 0.000 |
| Repeated subs. (B3) | yes | 0.953 | 0.957 | 0.947 | 0.948 | 1.447 | 0.553 | 0.130 | 0.000 |
| GWCP (B4) | no | 0.949 | 0.956 | 0.934 | 0.949 | 1.442 | 0.558 | 0.128 | 0.000 |

*All methods at $\alpha = 0.05$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| GWCP (B4) | yes | 0.954 | 0.957 | 0.948 | 0.949 | 1.451 | 0.549 | 0.128 | 0.000 |
| ProtCOVER | no | 0.954 | 0.960 | 0.942 | 0.953 | 1.469 | 0.531 | 0.111 | 0.000 |
| ProtCOVER | yes | 0.959 | 0.961 | 0.956 | 0.953 | 1.478 | 0.522 | 0.109 | 0.000 |
| ProtCOVER-D | no | 0.955 | 0.952 | 0.929 | 0.960 | 1.425 | 0.575 | 0.117 | 0.000 |
| ProtCOVER-D | yes | 0.958 | 0.960 | 0.955 | 0.955 | 1.484 | 0.516 | 0.117 | 0.000 |
| ProtCOVER-W | no | 0.953 | 0.960 | 0.939 | 0.952 | 1.462 | 0.538 | 0.113 | 0.000 |
| ProtCOVER-W | yes | 0.957 | 0.960 | 0.954 | 0.951 | 1.474 | 0.526 | 0.112 | 0.000 |
| (i)+(iii) gate | no | 0.949 | 0.956 | 0.933 | 0.949 | 1.441 | 0.559 | 0.127 | 0.000 |
| (i)+(iii) gate | yes | 0.953 | 0.957 | 0.947 | 0.948 | 1.447 | 0.553 | 0.130 | 0.000 |
| ProtCOVER-W + gate | no | 0.953 | 0.960 | 0.939 | 0.952 | 1.462 | 0.538 | 0.113 | 0.000 |
| ProtCOVER-W + gate | yes | 0.957 | 0.960 | 0.954 | 0.951 | 1.474 | 0.526 | 0.112 | 0.000 |
| Support-only (B5) | no | 0.953 | 0.960 | 0.939 | 0.952 | 1.462 | 0.538 | 0.113 | 0.000 |
| Support-only (B5) | yes | 0.957 | 0.960 | 0.954 | 0.951 | 1.474 | 0.526 | 0.112 | 0.000 |
| Random abst. (B5) | no | 0.953 | 0.960 | 0.939 | 0.952 | 1.462 | 0.538 | 0.113 | 0.000 |
| Random abst. (B5) | yes | 0.957 | 0.960 | 0.954 | 0.951 | 1.474 | 0.526 | 0.112 | 0.000 |
| *S2*, $\beta = 1$ | | | | | | | | | |
| Split (B1/B2) | no | 0.937 | 0.945 | 0.913 | 0.937 | 1.364 | 0.636 | 0.167 | 0.000 |
| Split (B1/B2) | yes | 0.939 | 0.944 | 0.902 | 0.945 | 1.361 | 0.639 | 0.165 | 0.000 |
| HCP (B3) | no | 0.947 | 0.955 | 0.929 | 0.949 | 1.439 | 0.561 | 0.133 | 0.000 |
| HCP (B3) | yes | 0.952 | 0.955 | 0.941 | 0.948 | 1.442 | 0.558 | 0.134 | 0.000 |
| Subsample once (B3) | no | 0.948 | 0.955 | 0.931 | 0.948 | 1.445 | 0.555 | 0.134 | 0.000 |
| Subsample once (B3) | yes | 0.950 | 0.953 | 0.939 | 0.946 | 1.434 | 0.566 | 0.142 | 0.000 |
| Repeated subs. (B3) | no | 0.947 | 0.955 | 0.930 | 0.949 | 1.440 | 0.560 | 0.134 | 0.000 |
| Repeated subs. (B3) | yes | 0.952 | 0.955 | 0.941 | 0.948 | 1.441 | 0.559 | 0.135 | 0.000 |
| GWCP (B4) | no | 0.948 | 0.956 | 0.932 | 0.949 | 1.448 | 0.552 | 0.129 | 0.000 |
| GWCP (B4) | yes | 0.954 | 0.957 | 0.946 | 0.950 | 1.455 | 0.545 | 0.129 | 0.000 |
| ProtCOVER | no | 0.958 | 0.963 | 0.947 | 0.958 | 1.502 | 0.498 | 0.102 | 0.000 |
| ProtCOVER | yes | 0.961 | 0.962 | 0.956 | 0.957 | 1.499 | 0.501 | 0.104 | 0.000 |
| ProtCOVER-D | no | 0.955 | 0.952 | 0.925 | 0.962 | 1.429 | 0.571 | 0.118 | 0.000 |
| ProtCOVER-D | yes | 0.959 | 0.961 | 0.955 | 0.958 | 1.496 | 0.504 | 0.113 | 0.000 |
| ProtCOVER-W | no | 0.955 | 0.961 | 0.940 | 0.955 | 1.486 | 0.514 | 0.107 | 0.000 |
| ProtCOVER-W | yes | 0.959 | 0.961 | 0.953 | 0.955 | 1.485 | 0.515 | 0.106 | 0.000 |
| (i)+(iii) gate | no | 0.947 | 0.955 | 0.929 | 0.949 | 1.439 | 0.561 | 0.133 | 0.000 |
| (i)+(iii) gate | yes | 0.952 | 0.955 | 0.941 | 0.948 | 1.442 | 0.558 | 0.134 | 0.000 |
| ProtCOVER-W + gate | no | 0.955 | 0.961 | 0.940 | 0.955 | 1.486 | 0.514 | 0.107 | 0.000 |
| ProtCOVER-W + gate | yes | 0.959 | 0.961 | 0.953 | 0.955 | 1.485 | 0.515 | 0.106 | 0.000 |
| Support-only (B5) | no | 0.955 | 0.961 | 0.940 | 0.955 | 1.486 | 0.514 | 0.107 | 0.000 |
| Support-only (B5) | yes | 0.959 | 0.961 | 0.953 | 0.955 | 1.485 | 0.515 | 0.106 | 0.000 |
| Random abst. (B5) | no | 0.955 | 0.961 | 0.940 | 0.955 | 1.486 | 0.514 | 0.107 | 0.000 |
| Random abst. (B5) | yes | 0.959 | 0.961 | 0.953 | 0.955 | 1.485 | 0.515 | 0.106 | 0.000 |
| *S2*, $\beta = 2$ | | | | | | | | | |
| Split (B1/B2) | no | 0.935 | 0.943 | 0.908 | 0.936 | 1.361 | 0.639 | 0.175 | 0.000 |
| Split (B1/B2) | yes | 0.937 | 0.942 | 0.895 | 0.944 | 1.357 | 0.643 | 0.173 | 0.000 |
| HCP (B3) | no | 0.946 | 0.953 | 0.926 | 0.947 | 1.435 | 0.565 | 0.140 | 0.000 |
| HCP (B3) | yes | 0.948 | 0.952 | 0.934 | 0.945 | 1.431 | 0.569 | 0.147 | 0.000 |
| Subsample once (B3) | no | 0.944 | 0.950 | 0.920 | 0.947 | 1.418 | 0.582 | 0.148 | 0.000 |
| Subsample once (B3) | yes | 0.947 | 0.951 | 0.934 | 0.944 | 1.428 | 0.572 | 0.151 | 0.000 |
| Repeated subs. (B3) | no | 0.946 | 0.953 | 0.926 | 0.947 | 1.435 | 0.565 | 0.140 | 0.000 |
| Repeated subs. (B3) | yes | 0.948 | 0.952 | 0.934 | 0.945 | 1.431 | 0.569 | 0.147 | 0.000 |

*All methods at $\alpha = 0.05$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| GWCP (B4) | no | 0.947 | 0.954 | 0.929 | 0.948 | 1.452 | 0.548 | 0.139 | 0.000 |
| GWCP (B4) | yes | 0.948 | 0.951 | 0.943 | 0.942 | 1.448 | 0.552 | 0.157 | 0.000 |
| ProtCOVER | no | 0.969 | 0.972 | 0.962 | 0.968 | 1.625 | 0.375 | 0.074 | 0.015 |
| ProtCOVER | yes | 0.971 | 0.972 | 0.969 | 0.967 | 1.621 | 0.379 | 0.079 | 0.015 |
| ProtCOVER-D | no | 0.955 | 0.953 | 0.924 | 0.961 | 1.437 | 0.563 | 0.121 | 0.000 |
| ProtCOVER-D | yes | 0.959 | 0.960 | 0.955 | 0.956 | 1.513 | 0.487 | 0.117 | 0.000 |
| ProtCOVER-W | no | 0.962 | 0.967 | 0.952 | 0.961 | 1.549 | 0.451 | 0.090 | 0.000 |
| ProtCOVER-W | yes | 0.964 | 0.966 | 0.962 | 0.959 | 1.541 | 0.459 | 0.094 | 0.000 |
| (i)+(iii) gate | no | 0.946 | 0.953 | 0.926 | 0.947 | 1.435 | 0.565 | 0.140 | 0.000 |
| (i)+(iii) gate | yes | 0.948 | 0.952 | 0.934 | 0.945 | 1.431 | 0.569 | 0.147 | 0.000 |
| ProtCOVER-W + gate | no | 0.962 | 0.967 | 0.952 | 0.961 | 1.549 | 0.451 | 0.090 | 0.000 |
| ProtCOVER-W + gate | yes | 0.964 | 0.966 | 0.962 | 0.959 | 1.543 | 0.457 | 0.094 | 0.007 |
| Support-only (B5) | no | 0.962 | 0.967 | 0.952 | 0.961 | 1.549 | 0.451 | 0.090 | 0.000 |
| Support-only (B5) | yes | 0.964 | 0.966 | 0.962 | 0.959 | 1.543 | 0.457 | 0.094 | 0.007 |
| Random abst. (B5) | no | 0.962 | 0.967 | 0.952 | 0.961 | 1.549 | 0.451 | 0.090 | 0.000 |
| Random abst. (B5) | yes | 0.964 | 0.966 | 0.962 | 0.959 | 1.544 | 0.456 | 0.094 | 0.007 |
| *S3 (orphan families)* | | | | | | | | | |
| Split (B1/B2) | no | 0.938 | 0.946 | 0.924 | 0.935 | 1.371 | 0.629 | 0.154 | 0.000 |
| Split (B1/B2) | yes | 0.941 | 0.946 | 0.918 | 0.942 | 1.371 | 0.629 | 0.149 | 0.000 |
| HCP (B3) | no | 0.950 | 0.956 | 0.938 | 0.950 | 1.451 | 0.549 | 0.116 | 0.000 |
| HCP (B3) | yes | 0.956 | 0.959 | 0.954 | 0.949 | 1.459 | 0.541 | 0.117 | 0.000 |
| Subsample once (B3) | no | 0.948 | 0.954 | 0.936 | 0.947 | 1.442 | 0.558 | 0.128 | 0.000 |
| Subsample once (B3) | yes | 0.955 | 0.957 | 0.953 | 0.948 | 1.454 | 0.546 | 0.118 | 0.000 |
| Repeated subs. (B3) | no | 0.950 | 0.955 | 0.937 | 0.950 | 1.449 | 0.551 | 0.116 | 0.000 |
| Repeated subs. (B3) | yes | 0.956 | 0.959 | 0.955 | 0.950 | 1.459 | 0.541 | 0.116 | 0.000 |
| GWCP (B4) | no | 0.951 | 0.956 | 0.938 | 0.953 | 1.461 | 0.539 | 0.117 | 0.000 |
| GWCP (B4) | yes | 0.956 | 0.957 | 0.949 | 0.952 | 1.448 | 0.552 | 0.118 | 0.000 |
| ProtCOVER | no | 0.956 | 0.960 | 0.943 | 0.958 | 1.486 | 0.514 | 0.101 | 0.000 |
| ProtCOVER | yes | 0.960 | 0.961 | 0.953 | 0.958 | 1.477 | 0.523 | 0.099 | 0.000 |
| ProtCOVER-D | no | 0.958 | 0.956 | 0.935 | 0.963 | 1.438 | 0.562 | 0.109 | 0.000 |
| ProtCOVER-D | yes | 0.959 | 0.959 | 0.955 | 0.958 | 1.461 | 0.539 | 0.111 | 0.000 |
| ProtCOVER-W | no | 0.956 | 0.959 | 0.942 | 0.958 | 1.485 | 0.515 | 0.104 | 0.000 |
| ProtCOVER-W | yes | 0.960 | 0.961 | 0.951 | 0.959 | 1.473 | 0.527 | 0.100 | 0.000 |
| (i)+(iii) gate | no | 0.950 | 0.956 | 0.938 | 0.950 | 1.451 | 0.549 | 0.116 | 0.000 |
| (i)+(iii) gate | yes | 0.956 | 0.959 | 0.955 | 0.950 | 1.460 | 0.540 | 0.116 | 0.003 |
| ProtCOVER-W + gate | no | 0.956 | 0.959 | 0.942 | 0.958 | 1.485 | 0.515 | 0.104 | 0.000 |
| ProtCOVER-W + gate | yes | 0.960 | 0.961 | 0.951 | 0.959 | 1.473 | 0.527 | 0.100 | 0.001 |
| Support-only (B5) | no | 0.956 | 0.959 | 0.942 | 0.958 | 1.485 | 0.515 | 0.104 | 0.000 |
| Support-only (B5) | yes | 0.960 | 0.961 | 0.951 | 0.959 | 1.473 | 0.527 | 0.100 | 0.001 |
| Random abst. (B5) | no | 0.956 | 0.959 | 0.942 | 0.958 | 1.485 | 0.515 | 0.104 | 0.000 |
| Random abst. (B5) | yes | 0.960 | 0.961 | 0.951 | 0.959 | 1.473 | 0.527 | 0.100 | 0.001 |
| *S4 (held-out regions)* | | | | | | | | | |
| Split (B1/B2) | no | 0.939 | 0.947 | 0.925 | 0.938 | 1.377 | 0.623 | 0.152 | 0.000 |
| Split (B1/B2) | yes | 0.941 | 0.947 | 0.922 | 0.942 | 1.375 | 0.625 | 0.145 | 0.000 |
| HCP (B3) | no | 0.949 | 0.956 | 0.936 | 0.951 | 1.444 | 0.556 | 0.122 | 0.000 |
| HCP (B3) | yes | 0.954 | 0.958 | 0.951 | 0.948 | 1.446 | 0.554 | 0.118 | 0.000 |
| Subsample once (B3) | no | 0.948 | 0.955 | 0.936 | 0.949 | 1.443 | 0.557 | 0.125 | 0.000 |
| Subsample once (B3) | yes | 0.953 | 0.958 | 0.951 | 0.947 | 1.442 | 0.558 | 0.122 | 0.000 |
| Repeated subs. (B3) | no | 0.949 | 0.956 | 0.937 | 0.950 | 1.445 | 0.555 | 0.122 | 0.000 |

*All methods at $\alpha = 0.05$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| Repeated subs. (B3) | yes | 0.954 | 0.958 | 0.951 | 0.948 | 1.445 | 0.555 | 0.119 | 0.000 |
| GWCP (B4) | no | 0.949 | 0.955 | 0.936 | 0.950 | 1.440 | 0.560 | 0.127 | 0.000 |
| GWCP (B4) | yes | 0.953 | 0.956 | 0.948 | 0.948 | 1.436 | 0.564 | 0.125 | 0.000 |
| ProtCOVER | no | 0.956 | 0.961 | 0.944 | 0.958 | 1.495 | 0.505 | 0.107 | 0.028 |
| ProtCOVER | yes | 0.960 | 0.962 | 0.958 | 0.955 | 1.495 | 0.505 | 0.108 | 0.028 |
| ProtCOVER-D | no | 0.958 | 0.956 | 0.936 | 0.964 | 1.458 | 0.542 | 0.106 | 0.000 |
| ProtCOVER-D | yes | 0.958 | 0.960 | 0.954 | 0.957 | 1.481 | 0.519 | 0.117 | 0.000 |
| ProtCOVER-W | no | 0.954 | 0.959 | 0.941 | 0.957 | 1.485 | 0.515 | 0.111 | 0.000 |
| ProtCOVER-W | yes | 0.956 | 0.957 | 0.949 | 0.953 | 1.465 | 0.535 | 0.122 | 0.000 |
| (i)+(iii) gate | no | 0.949 | 0.956 | 0.936 | 0.951 | 1.444 | 0.556 | 0.122 | 0.000 |
| (i)+(iii) gate | yes | 0.954 | 0.958 | 0.952 | 0.948 | 1.448 | 0.552 | 0.118 | 0.002 |
| ProtCOVER-W + gate | no | 0.954 | 0.959 | 0.941 | 0.957 | 1.485 | 0.515 | 0.111 | 0.000 |
| ProtCOVER-W + gate | yes | 0.956 | 0.957 | 0.949 | 0.953 | 1.465 | 0.535 | 0.122 | 0.002 |
| Support-only (B5) | no | 0.954 | 0.959 | 0.941 | 0.957 | 1.485 | 0.515 | 0.111 | 0.000 |
| Support-only (B5) | yes | 0.956 | 0.957 | 0.949 | 0.953 | 1.465 | 0.535 | 0.122 | 0.002 |
| Random abst. (B5) | no | 0.954 | 0.959 | 0.941 | 0.957 | 1.485 | 0.515 | 0.111 | 0.000 |
| Random abst. (B5) | yes | 0.956 | 0.957 | 0.949 | 0.953 | 1.465 | 0.535 | 0.122 | 0.002 |

Table 13: All methods at $\alpha = 0.2$, mean over predictors and seeds.

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| *S1 (exchangeable)* | | | | | | | | | |
| Split (B1/B2) | no | 0.797 | 0.794 | 0.753 | 0.804 | 0.954 | 0.934 | 0.280 | 0.000 |
| Split (B1/B2) | yes | 0.791 | 0.793 | 0.758 | 0.792 | 0.947 | 0.938 | 0.290 | 0.000 |
| HCP (B3) | no | 0.799 | 0.797 | 0.755 | 0.807 | 0.958 | 0.943 | 0.272 | 0.000 |
| HCP (B3) | yes | 0.812 | 0.813 | 0.800 | 0.800 | 0.994 | 0.939 | 0.246 | 0.000 |
| Subsample once (B3) | no | 0.799 | 0.797 | 0.753 | 0.808 | 0.959 | 0.938 | 0.273 | 0.000 |
| Subsample once (B3) | yes | 0.812 | 0.812 | 0.799 | 0.801 | 0.994 | 0.937 | 0.251 | 0.000 |
| Repeated subs. (B3) | no | 0.799 | 0.797 | 0.755 | 0.807 | 0.958 | 0.943 | 0.271 | 0.000 |
| Repeated subs. (B3) | yes | 0.812 | 0.813 | 0.800 | 0.800 | 0.994 | 0.939 | 0.245 | 0.000 |
| GWCP (B4) | no | 0.798 | 0.795 | 0.754 | 0.805 | 0.955 | 0.942 | 0.276 | 0.000 |
| GWCP (B4) | yes | 0.811 | 0.812 | 0.800 | 0.799 | 0.992 | 0.938 | 0.249 | 0.000 |
| ProtCOVER | no | 0.802 | 0.796 | 0.757 | 0.810 | 0.956 | 0.941 | 0.269 | 0.000 |
| ProtCOVER | yes | 0.812 | 0.813 | 0.802 | 0.803 | 0.993 | 0.941 | 0.251 | 0.000 |
| ProtCOVER-D | no | 0.803 | 0.793 | 0.755 | 0.815 | 0.946 | 0.929 | 0.273 | 0.000 |
| ProtCOVER-D | yes | 0.808 | 0.810 | 0.806 | 0.803 | 0.981 | 0.947 | 0.264 | 0.000 |
| ProtCOVER-W | no | 0.800 | 0.798 | 0.756 | 0.807 | 0.959 | 0.945 | 0.270 | 0.000 |
| ProtCOVER-W | yes | 0.812 | 0.813 | 0.799 | 0.801 | 0.994 | 0.937 | 0.247 | 0.000 |
| (i)+(iii) gate | no | 0.799 | 0.797 | 0.755 | 0.807 | 0.958 | 0.943 | 0.272 | 0.000 |
| (i)+(iii) gate | yes | 0.812 | 0.813 | 0.800 | 0.800 | 0.994 | 0.939 | 0.246 | 0.000 |
| ProtCOVER-W + gate | no | 0.800 | 0.798 | 0.756 | 0.808 | 0.959 | 0.944 | 0.269 | 0.000 |
| ProtCOVER-W + gate | yes | 0.812 | 0.813 | 0.799 | 0.802 | 0.995 | 0.936 | 0.247 | 0.002 |
| Support-only (B5) | no | 0.800 | 0.798 | 0.756 | 0.807 | 0.959 | 0.944 | 0.270 | 0.000 |
| Support-only (B5) | yes | 0.812 | 0.813 | 0.799 | 0.802 | 0.995 | 0.936 | 0.247 | 0.002 |
| Random abst. (B5) | no | 0.800 | 0.798 | 0.756 | 0.807 | 0.959 | 0.944 | 0.270 | 0.000 |
| Random abst. (B5) | yes | 0.812 | 0.813 | 0.799 | 0.802 | 0.996 | 0.936 | 0.247 | 0.002 |
| *S2, $\beta = 0.5$* | | | | | | | | | |
| Split (B1/B2) | no | 0.792 | 0.791 | 0.741 | 0.802 | 0.950 | 0.928 | 0.294 | 0.000 |

*All methods at $\alpha = 0.2$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| Split (B1/B2) | yes | 0.787 | 0.789 | 0.741 | 0.794 | 0.944 | 0.933 | 0.304 | 0.000 |
| HCP (B3) | no | 0.794 | 0.792 | 0.742 | 0.804 | 0.952 | 0.941 | 0.289 | 0.000 |
| HCP (B3) | yes | 0.805 | 0.804 | 0.783 | 0.798 | 0.982 | 0.937 | 0.267 | 0.000 |
| Subsample once (B3) | no | 0.791 | 0.789 | 0.737 | 0.803 | 0.946 | 0.937 | 0.298 | 0.000 |
| Subsample once (B3) | yes | 0.804 | 0.803 | 0.783 | 0.797 | 0.982 | 0.937 | 0.271 | 0.000 |
| Repeated subs. (B3) | no | 0.794 | 0.792 | 0.743 | 0.804 | 0.952 | 0.942 | 0.289 | 0.000 |
| Repeated subs. (B3) | yes | 0.805 | 0.804 | 0.782 | 0.798 | 0.981 | 0.937 | 0.268 | 0.000 |
| GWCP (B4) | no | 0.795 | 0.793 | 0.743 | 0.805 | 0.954 | 0.943 | 0.285 | 0.000 |
| GWCP (B4) | yes | 0.809 | 0.811 | 0.794 | 0.798 | 0.995 | 0.937 | 0.258 | 0.000 |
| ProtCOVER | no | 0.798 | 0.796 | 0.749 | 0.808 | 0.960 | 0.944 | 0.278 | 0.000 |
| ProtCOVER | yes | 0.810 | 0.807 | 0.799 | 0.801 | 0.997 | 0.931 | 0.259 | 0.000 |
| ProtCOVER-D | no | 0.801 | 0.792 | 0.745 | 0.814 | 0.949 | 0.932 | 0.280 | 0.000 |
| ProtCOVER-D | yes | 0.807 | 0.810 | 0.801 | 0.804 | 0.988 | 0.949 | 0.266 | 0.000 |
| ProtCOVER-W | no | 0.797 | 0.796 | 0.746 | 0.807 | 0.959 | 0.946 | 0.280 | 0.000 |
| ProtCOVER-W | yes | 0.813 | 0.815 | 0.801 | 0.801 | 1.004 | 0.939 | 0.251 | 0.000 |
| (i)+(iii) gate | no | 0.794 | 0.792 | 0.742 | 0.804 | 0.952 | 0.941 | 0.289 | 0.000 |
| (i)+(iii) gate | yes | 0.805 | 0.804 | 0.783 | 0.799 | 0.982 | 0.937 | 0.267 | 0.000 |
| ProtCOVER-W + gate | no | 0.797 | 0.796 | 0.746 | 0.807 | 0.959 | 0.945 | 0.279 | 0.000 |
| ProtCOVER-W + gate | yes | 0.813 | 0.815 | 0.801 | 0.801 | 1.005 | 0.938 | 0.250 | 0.001 |
| Support-only (B5) | no | 0.797 | 0.796 | 0.746 | 0.807 | 0.959 | 0.945 | 0.279 | 0.000 |
| Support-only (B5) | yes | 0.813 | 0.815 | 0.801 | 0.801 | 1.005 | 0.937 | 0.250 | 0.001 |
| Random abst. (B5) | no | 0.797 | 0.796 | 0.746 | 0.807 | 0.959 | 0.945 | 0.280 | 0.000 |
| Random abst. (B5) | yes | 0.813 | 0.815 | 0.801 | 0.801 | 1.005 | 0.937 | 0.250 | 0.001 |
| *S2*, $\beta = 1$ | | | | | | | | | |
| Split (B1/B2) | no | 0.790 | 0.788 | 0.732 | 0.801 | 0.947 | 0.924 | 0.307 | 0.000 |
| Split (B1/B2) | yes | 0.784 | 0.785 | 0.728 | 0.795 | 0.942 | 0.929 | 0.313 | 0.000 |
| HCP (B3) | no | 0.791 | 0.789 | 0.733 | 0.803 | 0.949 | 0.938 | 0.302 | 0.000 |
| HCP (B3) | yes | 0.800 | 0.798 | 0.766 | 0.799 | 0.973 | 0.933 | 0.285 | 0.000 |
| Subsample once (B3) | no | 0.786 | 0.784 | 0.726 | 0.798 | 0.940 | 0.930 | 0.317 | 0.000 |
| Subsample once (B3) | yes | 0.799 | 0.797 | 0.764 | 0.798 | 0.971 | 0.933 | 0.287 | 0.000 |
| Repeated subs. (B3) | no | 0.791 | 0.789 | 0.733 | 0.803 | 0.949 | 0.938 | 0.301 | 0.000 |
| Repeated subs. (B3) | yes | 0.800 | 0.797 | 0.766 | 0.798 | 0.973 | 0.933 | 0.286 | 0.000 |
| GWCP (B4) | no | 0.793 | 0.792 | 0.735 | 0.805 | 0.953 | 0.941 | 0.294 | 0.000 |
| GWCP (B4) | yes | 0.807 | 0.808 | 0.787 | 0.798 | 0.995 | 0.935 | 0.268 | 0.000 |
| ProtCOVER | no | 0.804 | 0.801 | 0.751 | 0.815 | 0.976 | 0.935 | 0.270 | 0.000 |
| ProtCOVER | yes | 0.818 | 0.813 | 0.802 | 0.810 | 1.020 | 0.912 | 0.248 | 0.000 |
| ProtCOVER-D | no | 0.799 | 0.792 | 0.738 | 0.814 | 0.952 | 0.933 | 0.286 | 0.000 |
| ProtCOVER-D | yes | 0.808 | 0.810 | 0.794 | 0.807 | 0.992 | 0.951 | 0.268 | 0.000 |
| ProtCOVER-W | no | 0.800 | 0.798 | 0.745 | 0.811 | 0.966 | 0.949 | 0.275 | 0.000 |
| ProtCOVER-W | yes | 0.816 | 0.817 | 0.800 | 0.805 | 1.015 | 0.938 | 0.245 | 0.000 |
| (i)+(iii) gate | no | 0.791 | 0.789 | 0.733 | 0.803 | 0.949 | 0.938 | 0.302 | 0.000 |
| (i)+(iii) gate | yes | 0.800 | 0.798 | 0.766 | 0.799 | 0.974 | 0.933 | 0.285 | 0.000 |
| ProtCOVER-W + gate | no | 0.800 | 0.798 | 0.745 | 0.811 | 0.966 | 0.949 | 0.274 | 0.000 |
| ProtCOVER-W + gate | yes | 0.816 | 0.818 | 0.801 | 0.805 | 1.016 | 0.937 | 0.244 | 0.001 |
| Support-only (B5) | no | 0.800 | 0.798 | 0.745 | 0.811 | 0.966 | 0.949 | 0.274 | 0.000 |
| Support-only (B5) | yes | 0.816 | 0.818 | 0.801 | 0.805 | 1.016 | 0.937 | 0.244 | 0.001 |
| Random abst. (B5) | no | 0.800 | 0.798 | 0.745 | 0.811 | 0.966 | 0.949 | 0.274 | 0.000 |
| Random abst. (B5) | yes | 0.816 | 0.818 | 0.801 | 0.805 | 1.016 | 0.937 | 0.244 | 0.001 |

*All methods at $\alpha = 0.2$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| *S2, $\beta = 2$* | | | | | | | | | |
| Split (B1/B2) | no | 0.789 | 0.787 | 0.726 | 0.801 | 0.947 | 0.922 | 0.309 | 0.000 |
| Split (B1/B2) | yes | 0.782 | 0.784 | 0.722 | 0.793 | 0.940 | 0.926 | 0.318 | 0.000 |
| HCP (B3) | no | 0.790 | 0.788 | 0.727 | 0.802 | 0.947 | 0.936 | 0.305 | 0.000 |
| HCP (B3) | yes | 0.795 | 0.793 | 0.757 | 0.795 | 0.966 | 0.931 | 0.300 | 0.000 |
| Subsample once (B3) | no | 0.789 | 0.786 | 0.723 | 0.802 | 0.944 | 0.935 | 0.308 | 0.000 |
| Subsample once (B3) | yes | 0.795 | 0.792 | 0.755 | 0.795 | 0.966 | 0.930 | 0.301 | 0.000 |
| Repeated subs. (B3) | no | 0.790 | 0.788 | 0.727 | 0.802 | 0.947 | 0.936 | 0.305 | 0.000 |
| Repeated subs. (B3) | yes | 0.795 | 0.793 | 0.757 | 0.795 | 0.966 | 0.931 | 0.300 | 0.000 |
| GWCP (B4) | no | 0.796 | 0.794 | 0.733 | 0.808 | 0.959 | 0.935 | 0.291 | 0.000 |
| GWCP (B4) | yes | 0.803 | 0.805 | 0.779 | 0.794 | 0.992 | 0.925 | 0.283 | 0.000 |
| ProtCOVER | no | 0.829 | 0.826 | 0.785 | 0.836 | 1.064 | 0.862 | 0.224 | 0.000 |
| ProtCOVER | yes | 0.835 | 0.830 | 0.832 | 0.821 | 1.114 | 0.826 | 0.221 | 0.000 |
| ProtCOVER-D | no | 0.799 | 0.795 | 0.733 | 0.814 | 0.959 | 0.936 | 0.285 | 0.000 |
| ProtCOVER-D | yes | 0.805 | 0.808 | 0.783 | 0.807 | 0.998 | 0.947 | 0.274 | 0.000 |
| ProtCOVER-W | no | 0.806 | 0.804 | 0.748 | 0.817 | 0.978 | 0.942 | 0.263 | 0.000 |
| ProtCOVER-W | yes | 0.815 | 0.815 | 0.796 | 0.805 | 1.015 | 0.928 | 0.253 | 0.000 |
| (i)+(iii) gate | no | 0.790 | 0.788 | 0.727 | 0.802 | 0.948 | 0.936 | 0.305 | 0.000 |
| (i)+(iii) gate | yes | 0.796 | 0.793 | 0.757 | 0.795 | 0.966 | 0.931 | 0.300 | 0.000 |
| ProtCOVER-W + gate | no | 0.812 | 0.810 | 0.755 | 0.822 | 1.010 | 0.912 | 0.254 | 0.030 |
| ProtCOVER-W + gate | yes | 0.820 | 0.820 | 0.801 | 0.810 | 1.040 | 0.903 | 0.247 | 0.030 |
| Support-only (B5) | no | 0.812 | 0.809 | 0.754 | 0.822 | 1.005 | 0.918 | 0.256 | 0.030 |
| Support-only (B5) | yes | 0.820 | 0.819 | 0.800 | 0.810 | 1.040 | 0.903 | 0.248 | 0.030 |
| Random abst. (B5) | no | 0.812 | 0.810 | 0.755 | 0.822 | 1.010 | 0.914 | 0.254 | 0.030 |
| Random abst. (B5) | yes | 0.820 | 0.820 | 0.800 | 0.810 | 1.041 | 0.901 | 0.247 | 0.030 |
| *S3 (orphan families)* | | | | | | | | | |
| Split (B1/B2) | no | 0.794 | 0.793 | 0.750 | 0.801 | 0.952 | 0.927 | 0.295 | 0.000 |
| Split (B1/B2) | yes | 0.787 | 0.789 | 0.754 | 0.787 | 0.943 | 0.934 | 0.308 | 0.000 |
| HCP (B3) | no | 0.797 | 0.796 | 0.753 | 0.804 | 0.956 | 0.938 | 0.284 | 0.000 |
| HCP (B3) | yes | 0.811 | 0.813 | 0.803 | 0.795 | 0.989 | 0.940 | 0.259 | 0.000 |
| Subsample once (B3) | no | 0.798 | 0.796 | 0.750 | 0.809 | 0.959 | 0.938 | 0.282 | 0.000 |
| Subsample once (B3) | yes | 0.811 | 0.813 | 0.804 | 0.795 | 0.991 | 0.940 | 0.259 | 0.000 |
| Repeated subs. (B3) | no | 0.797 | 0.796 | 0.753 | 0.804 | 0.956 | 0.938 | 0.285 | 0.000 |
| Repeated subs. (B3) | yes | 0.811 | 0.813 | 0.803 | 0.795 | 0.989 | 0.940 | 0.258 | 0.000 |
| GWCP (B4) | no | 0.803 | 0.802 | 0.760 | 0.810 | 0.967 | 0.945 | 0.266 | 0.000 |
| GWCP (B4) | yes | 0.814 | 0.814 | 0.800 | 0.804 | 0.993 | 0.941 | 0.250 | 0.000 |
| ProtCOVER | no | 0.806 | 0.805 | 0.764 | 0.813 | 0.974 | 0.942 | 0.260 | 0.000 |
| ProtCOVER | yes | 0.818 | 0.817 | 0.798 | 0.812 | 1.001 | 0.941 | 0.244 | 0.000 |
| ProtCOVER-D | no | 0.810 | 0.803 | 0.762 | 0.820 | 0.967 | 0.940 | 0.264 | 0.000 |
| ProtCOVER-D | yes | 0.813 | 0.811 | 0.806 | 0.811 | 0.993 | 0.951 | 0.254 | 0.000 |
| ProtCOVER-W | no | 0.807 | 0.806 | 0.764 | 0.814 | 0.975 | 0.943 | 0.258 | 0.000 |
| ProtCOVER-W | yes | 0.819 | 0.817 | 0.796 | 0.813 | 1.000 | 0.942 | 0.242 | 0.000 |
| (i)+(iii) gate | no | 0.797 | 0.796 | 0.753 | 0.804 | 0.956 | 0.938 | 0.284 | 0.000 |
| (i)+(iii) gate | yes | 0.811 | 0.813 | 0.804 | 0.795 | 0.990 | 0.939 | 0.258 | 0.002 |
| ProtCOVER-W + gate | no | 0.807 | 0.806 | 0.765 | 0.814 | 0.977 | 0.942 | 0.257 | 0.002 |
| ProtCOVER-W + gate | yes | 0.819 | 0.817 | 0.797 | 0.814 | 1.003 | 0.939 | 0.241 | 0.004 |
| Support-only (B5) | no | 0.807 | 0.806 | 0.765 | 0.814 | 0.977 | 0.942 | 0.257 | 0.002 |
| Support-only (B5) | yes | 0.819 | 0.817 | 0.797 | 0.814 | 1.003 | 0.939 | 0.240 | 0.004 |
| Random abst. (B5) | no | 0.807 | 0.806 | 0.765 | 0.814 | 0.977 | 0.942 | 0.257 | 0.002 |

*All methods at $\alpha = 0.2$, continued*

| Method | cc | Cov | var.-w. | $\text{Cov}_1$ | $\text{Cov}_0$ | size | single. | under. | abst. |
|---|---|---|---|---|---|---|---|---|---|
| Random abst. (B5) | yes | 0.819 | 0.817 | 0.797 | 0.814 | 1.004 | 0.938 | 0.240 | 0.004 |
| *S4 (held-out regions)* | | | | | | | | | |
| Split (B1/B2) | no | 0.797 | 0.792 | 0.754 | 0.811 | 0.959 | 0.930 | 0.283 | 0.000 |
| Split (B1/B2) | yes | 0.791 | 0.793 | 0.763 | 0.792 | 0.951 | 0.937 | 0.294 | 0.000 |
| HCP (B3) | no | 0.796 | 0.790 | 0.753 | 0.810 | 0.955 | 0.941 | 0.280 | 0.000 |
| HCP (B3) | yes | 0.811 | 0.812 | 0.803 | 0.799 | 0.993 | 0.937 | 0.250 | 0.000 |
| Subsample once (B3) | no | 0.796 | 0.790 | 0.752 | 0.810 | 0.956 | 0.941 | 0.280 | 0.000 |
| Subsample once (B3) | yes | 0.810 | 0.812 | 0.804 | 0.797 | 0.992 | 0.936 | 0.252 | 0.000 |
| Repeated subs. (B3) | no | 0.796 | 0.790 | 0.752 | 0.810 | 0.955 | 0.941 | 0.280 | 0.000 |
| Repeated subs. (B3) | yes | 0.811 | 0.812 | 0.803 | 0.799 | 0.993 | 0.937 | 0.251 | 0.000 |
| GWCP (B4) | no | 0.798 | 0.792 | 0.754 | 0.812 | 0.959 | 0.944 | 0.275 | 0.000 |
| GWCP (B4) | yes | 0.810 | 0.809 | 0.798 | 0.803 | 0.989 | 0.937 | 0.253 | 0.000 |
| ProtCOVER | no | 0.807 | 0.800 | 0.765 | 0.820 | 0.978 | 0.936 | 0.261 | 0.001 |
| ProtCOVER | yes | 0.822 | 0.822 | 0.815 | 0.812 | 1.020 | 0.925 | 0.234 | 0.001 |
| ProtCOVER-D | no | 0.807 | 0.793 | 0.760 | 0.824 | 0.957 | 0.935 | 0.264 | 0.000 |
| ProtCOVER-D | yes | 0.808 | 0.810 | 0.798 | 0.812 | 0.984 | 0.946 | 0.263 | 0.000 |
| ProtCOVER-W | no | 0.808 | 0.803 | 0.766 | 0.820 | 0.979 | 0.948 | 0.251 | 0.000 |
| ProtCOVER-W | yes | 0.815 | 0.811 | 0.796 | 0.813 | 1.000 | 0.937 | 0.248 | 0.000 |
| (i)+(iii) gate | no | 0.797 | 0.791 | 0.753 | 0.810 | 0.958 | 0.938 | 0.279 | 0.002 |
| (i)+(iii) gate | yes | 0.812 | 0.813 | 0.804 | 0.800 | 1.001 | 0.930 | 0.249 | 0.004 |
| ProtCOVER-W + gate | no | 0.808 | 0.803 | 0.767 | 0.821 | 0.982 | 0.945 | 0.250 | 0.002 |
| ProtCOVER-W + gate | yes | 0.816 | 0.812 | 0.797 | 0.814 | 1.008 | 0.930 | 0.246 | 0.005 |
| Support-only (B5) | no | 0.808 | 0.803 | 0.767 | 0.821 | 0.981 | 0.947 | 0.250 | 0.002 |
| Support-only (B5) | yes | 0.816 | 0.812 | 0.797 | 0.814 | 1.005 | 0.933 | 0.245 | 0.005 |
| Random abst. (B5) | no | 0.808 | 0.803 | 0.767 | 0.821 | 0.981 | 0.946 | 0.250 | 0.002 |
| Random abst. (B5) | yes | 0.816 | 0.812 | 0.797 | 0.814 | 1.006 | 0.933 | 0.246 | 0.005 |